

The role of governmental venture capital in value creation for investee firms: Evidence from Chinese government guidance funds

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Abstract

We study the role of Chinese government guidance funds (GGFs) in value creation for investee firms. Using a sample of 2,855 firms that went public during the period of 2010-2021, we show that GGF-backed IPO firms had higher initial returns than non-VC-backed IPO firms and non-GGF VC-backed IPO firms. After decomposing the initial return into IPO underpricing and market overvaluation, we find that GGF-backed firms enjoyed higher overvaluation and lower underpricing than other firms. Consistent with investor sentiment and information asymmetry hypothesis, our results indicate that public investors value the benefits of political resources more than the costs of government interference associated with GGF sponsoring. However, GGF-backed firms did not outperform other-VC-backed firms when post-IPO long-term stock, operating and innovation performance is assessed. The divergence in the effects of GGFs observed in the financial and product markets reveals the complexity in evaluating the role of GGFs in value creation.

Keywords: venture capital, government guidance funds, IPO initial return, underpricing, market overvaluation, post-IPO performance

JEL classification: G24, G32, G41

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1. Introduction

Over the past decade, the venture capital (VC) market in China has evolved quickly despite its short history since 2000, becoming the second largest in the world (only behind the US) and increasingly attractive to global VCs.² As of 2022, 19,354 VC funds have been registered with the Asset Management Association of China (AMAC), with the asset under management reaching a total value of CNY 2.83 trillion (US\$ 0.39 trillion).³ One important booster for the growth of China's VC market is the direct participation of the state as the capital provider of VC funds. That is, the governments at various levels set up government guidance funds (GGFs) to provide funding for firms that are considered to be important for economic development and under-funded, analogous to the governmental venture capital (GVC) programs adopted in other major markets.⁴ To understand the specifics of the GGFs is imperative for both the academia and global VCs with an interest in entering the Chinese market, since the state plays a central role in capital allocation and economic growth in China (Allen et al., 2022; Colinnelli et al., 2022), and the importance of GGFs in China's VC market has grown rapidly since 2015.⁵ While numerous studies have examined the effectiveness of GVC programs on stimulating the growth of the entrepreneurial firms, the VC market and the local economy in the western countries, with no consensus arrived (e.g. Lerner, 2000; Avnimelech and Teubal, 2006; Cumming, 2007; Alperovych et al., 2015; Brander et al., 2015; Cumming et al., 2017), scant attention has been paid to China's VC market. Thus, to fill this gap, this paper investigates whether and how GGFs create value for their investee firms in the financial market and the product market.

GGFs have three distinguished features in comparison with other types of VC funds. First, most of the GGFs rely on poorly incentivized governmental agencies or state-owned firms to deliver fund management, and even when GGFs cooperate with professional private fund

² See <https://kpmg.com/xx/en/home/campaigns/2021/04/q1-venture-pulse-report-asia.html>.

³ See <https://www.amac.org.cn/researchstatistics/report/zgsmjjhysjbg/>.

⁴ For instance, the Small Business Innovation Research Program (SBIR) in U.S., the Innovation Investment Fund (IIF) governmental program in Australia, the Labor Sponsored Venture Capital Corporations (LSVCCs) in Canada, and the Yozma program in Israel etc.

⁵ Details are discussed in Section 2.

managers, it is common that the governments intervene the decision-making of fund managers, which may reduce management efficiency (Lin, 2021). Second, GGFs have a dual mandate to generate financial returns and to support the government's policy goals (Luong et al., 2021). The distracted profit motives of GGFs may distort the capital allocation and reduce the motivation of VCs to conduct costly coaching and monitoring activities to spur the growth of the investee firms. Third, GGFs have natural ties with governments, which may bring valuable political resources to support the growth of their investee firms. Given these characteristics of GGFs, we ask the question of whether the benefits or the costs GGFs bring to the investee firms dominate in practice.

We address the question in the context of initial public offerings (IPO) which is the most important exit channel for the VCs in the Chinese market (Chen, 2022). Using a sample of 2,855 firms that went public during the period between 2010 and 2021 in China, we firstly show that the firms backed by GGFs (i.e. the firms receiving at least one investment from GGFs prior to IPOs), have higher IPO initial returns (i.e. the stock return at the first day post IPO) than the firms backed by non-GGF VCs (i.e. the firms receiving at least one investment from VCs without funding from GGFs prior to IPOs). Importantly, after decomposing the initial return into IPO underpricing (i.e. low offer price relative to firm true value) and market overvaluation (i.e. high first-day closing price relative to firm true value), we find that the GGF-backed firms enjoy lower underpricing and higher market overvaluation than the comparable non-GGF VC-backed firms. The results are robust after addressing the endogeneity concern by performing entropy balancing matching and Heckman two-step estimation, and are robust to alternative methods of estimating firms' true value.

The evidence, consistent with the information asymmetry theory, indicates that the public investors perceive the funding from GGFs as a positive signal for firm quality, which makes it less necessary for GGF-backed firms to depress the offer price to attract investors in the presence of information asymmetry at IPOs (e.g. Rock, 1986; Allen and Faulhaber, 1989; Banerjee et al., 2011; Buchner et al., 2019). Drawing on the investor sentiment theory, the evidence also indicates that public investors hold more optimistic attitudes towards the firms backed by GGFs in comparison with the firms backed by other VCs. Overall, the results imply

that investors value the benefits associated with political connections more than the possible costs associated with management inefficiency or capital misallocation brought by GGFs. This creates value not only for the IPO firms by reducing the ‘money left on the table’, but also for the investors by increasing the short-term financial returns they could earn from purchasing the new shares.

Heterogeneity analysis shows that investors’ evaluation towards GGF-backed firms varies across the stage and the amount GGFs invest in a firm. We conjecture that if a firm’s early VC investments are made by GGFs rather than market-oriented VCs, the market may regard it as an uncompetitive firm that lacks attractiveness to, and needs support from GGFs as an additional ‘selling point’ to attract attention from profit-seeking VCs. Also, we conjecture that the amount of GGF investments relative to other VCs a firm receives represents the degree of governmental intervention and management inefficiency, and the market may regard it as a negative signal of firm quality. We find evidence in support of the conjectures by showing that investors’ overvaluation towards the firms that receive GGF investments at an early stage, and that receive more GGF investments than other-VC investments is lower. In summary, the evidence indicates that investors’ optimistic sentiment towards GGF-backed firms can be diluted by the concern for the potential costs associated with the governmental intervention.

However, GGF-backed firms’ outperformance at IPOs (i.e. higher initial return) do not persist post IPOs. We show that although the stocks of non-GGF VC-backed firms, on average, outperform those of non-VC-backed firms by 0.5% per month over the two years post IPOs, GGF-backed firms do not show statistically significant differences in stock performance in comparison with non-VC-backed firms. This indicates that the high valuation for GGF-backed firms at IPOs arises from a temporary overvaluation by the market, rather than from real benefits GGFs could bring up to effectively improve firms’ long-term performance. Consistently, we also find that the return on assets of the GGF-backed firms in the two years following the IPOs is on average lower than that of the non-VC-backed firms by 0.6% per annum, whilst there is no statistically significant difference in the operating performance between the non-GGF VC-backed firms and non-VC-backed firms. The results are robust to alternative measures for operating performance. Moreover, we find no evidence that GGFs

differ from other VC funds in stimulating firms' innovation expenditures or innovation productivity. The examination of firms' performance in the product market reinforces the implication that the governmental intervention and management inefficiency coming along with the GGF capital may dominate the benefits of political connections when it comes to spurring the long-term growth of their investee firms.

Thus, this paper makes three important contributions to the literature. First, it adds to the literature evaluating the role of governmental participation in the VC market. Extant literature mainly focuses on the governmental venture capital programs carried out in the developed markets such as the US (Lerner, 2000), Europe (Beuselinck and Manigart, 2000; Balboa et al., 2007; Cumming and Johan, 2013; Grilli and Murtinu, 2014, 2015), Australia (Cumming, 2007; Cumming and Johan, 2013; Cumming et al., 2017), and Israel (Avnimelech and Teubal, 2006), and looks into the impact of these GVC programs on investee firms' product-market performance and on fostering the development of VC market and economic growth. This paper, to the best of our knowledge, is the first study evaluating the role of Chinese GGFs in value creation for the investee firms, and thus contributes to the literature by adding insights into China's VC market, which represents the second largest market for innovative and high-growth firms in the world (Colonnelli et al., 2022) and where the government plays a crucial role in capital allocation and shaping the business environment across the country (Claessens et al., 2008; Hsieh and Song, 2015; Jia et al., 2021). We show that the sponsoring by GGFs could create value in the financial market both for the IPO firms by lowering the 'money left on the table' and for the investors of purchasing the new shares by driving up the market overvaluation. However, the value created in the financial market is shown to be driven by temporary investor sentiment rather than the real benefits GGFs bring up to effectively improve the investee firms' performance in the product market. The ineffectiveness of GGFs in spurring the long-term growth of investee firms adds supportive evidence to the critiques of government efforts to promote VC finance (Lerner, 2010). In addition, the divergence of GGFs' value effects observed in the financial and the product market contributes to the literature by revealing the complexity of evaluating the value creation role of GGFs.

Second, the paper adds to the understanding of China's VC market. In comparison with

the extensive evidence documented for the VC markets in other countries (e.g. Kortum and Lerner, 2000; Hellmann and Puri, 2002; Puri and Zarutskie, 2012; Tian, 2011, 2012; Michala, 2019), China's VC market is still under-researched (Chen, 2022). Earlier studies discuss the institutional differences in the VC industry between China and the West (Brutona and Ahlstrom, 2003), the determinants of cross-border VC performance in China (Wang and Wang, 2011), and the effect of VC investments on the R&D activities and performance of portfolio firms in China (Guo and Jiang, 2013). However, there is a dearth of studies taking a more granular perspective and in particular assessing the role of governmental participation which is expected to have a prominent impact on China's VC market. One exception is the recent paper by Colinnelli et al. (2022) that surveys the preferences of private fund managers (i.e. GPs) on collaborating with capital providers (i.e. LPs) having governmental ties in China's venture capital and private equity market, concluding that the average fund manager disliked capital providers with governmental ties. Their findings indicate that professional fund managers hold a negative perception on governmental participation (i.e. the pros of political connections outweighing the cons of political interference in decision-making) in VC funding in the Chinese market. However, our paper reveals an opposite pattern in the IPO market which is dominated by unsophisticated retail investors; that is, the average public investor holds optimistic attitudes towards the IPO firms backed by VCs with governments as the capital providers. Thus, our paper complements the literature on China's VC market by identifying the perception of stock market investors towards governmental participation in VC funding. Also, the divergence between investors' bullish attitudes towards GGFs and the underperformance of GGFs in spurring long-term firm growth casts doubts on the ability of investors to evaluate new shares in the Chinese market.

Third, the paper contributes to the understanding of the impact of political connections on firms' IPO performance in the Chinese market. There is evidence that the firms that have politically-connected top executives, or that hire underwriters with political connections have a higher probability of obtaining IPO approvals (Chen et al., 2017; Wang and Wu, 2020). In particular, Wang and Wu (2020) find that firms backed by politically-connected VCs (i.e. the VCs having government officials as top executives) have higher initial returns at IPOs and

worse post-IPO performance than firms backed by non-politically-connected VCs. They argue that the firms backed by politically-connected VCs are less mature, of lower quality and demand more political resource in order to go public, which makes it necessary for firms to depress the offer price at IPOs. However, while we also document higher initial returns of firms backed by politically-connected VCs (i.e. GGF-backed firms), the decomposition of the IPO initial return into IPO underpricing and market overvaluation suggests that it is overvaluation rather than underpricing that drives the result. Thus, our evidence highlights the importance of disentangling underpricing from overvaluation when assessing IPO initial returns, in particular in the Chinese market where the retail investors whose trading behaviors tend to be driven by sentiment dominate.

The remainder of the paper proceeds as follows. Section 2 briefly describes the context of VC and IPO markets in China. Section 3 reviews relevant literature and develops testable hypotheses. Section 4 describes the sources of data. Section 5 describes variables, summary statistics and empirical methodology. Sections 6 and 7 report and discuss the results for the value creation role of GGFs in the financial market and in the product market, respectively. Section 8 concludes.

2. Institutional background

2.1 China's venture capital market and government guidance funds

China's venture capital market has been quickly evolving, in particular over the past decade, and became the second largest in the world (after the U.S.), despite its short history since 2000. Such a rapid growth can be largely attributed to the support from the Chinese government, either through indirect actions such as regulations or through direct actions such as being the capital supplier (Chen, 2022).

On the one hand, a batch of regulations have boosted the development of the VC market by stimulating the demand and the supply of the venture capital. On the demand side, the Chinese State Council launched the policy initiative "Mass Entrepreneurship and Innovation" in 2015, potentially increasing the supply of high-tech and high-growth entrepreneurship in

China and thus the demand for venture capital.⁶ On the supply side, for a long time, large financial institutions were not allowed to invest in China's VC market, and such a restriction started to loosen in recent years. For instance, insurance companies were allowed to invest in venture capital funds since 2014, indicating the government's intention to introduce sophisticated institutional investors and long-term capital providers to the VC market.⁷ The tax policy in 2017 and 2018 allows VC investors to deduct 70% of the invested capital from the taxable investment income to encourage funding for the venture capital.⁸ In addition, the increase in the exit channels for venture capital investors also contributes to the growth of the market. In 2009, 2019 and 2021, the Shenzhen ChiNext exchange, the Shanghai STAR exchange, and the Beijing Stock Exchange was launched in succession. The establishment of the new stock exchanges with lower listing criteria in comparison with the traditional ones (i.e. the Shanghai Stock Exchange and the Main board of the Shenzhen Stock Exchange) enriches entrepreneurial firms' access to the public equity market (Bernstein et al., 2020) and thus may attract funding from venture capital investors, in particular when the IPO serves as the most important exit channel for Chinese VC funds accounting for 60% of the exits (Chen, 2022).

On the other hand, Chinese governments at various levels directly set up a special type of VC funds, i.e. government guidance funds (GGFs) that are policy funds aimed at providing capital to and encouraging non-government investors (known as the 'social capital' from investment companies, wealthy families and individuals, and other VC/PE institutions etc.) to invest in firms that are perceived as crucial for economic and social development and under-funded.⁹ Most GGFs aim to raise 70 to 80 percent of the total funding from non-government investors (Zero2IPO, 2020). Such a design of high proportions of social capital in a GGF intends to bring profit motives, market discipline and professional expertise into industrial policy and alleviate the pitfalls of inefficiency and corruption associated with traditional industrial policy tools (Lin, 2021).

The first GGF, i.e. the Zhongguancun VC guidance fund was established in 2002, and the

⁶ See the initiative details at http://www.gov.cn/zhengce/content/2015-06/16/content_9855.htm (in Chinese).

⁷ See the regulation details at http://www.cbirc.gov.cn/cn/view/pages/ItemDetail_gdsj.html?docId=16472&docType=2 (in Chinese)

⁸ See the regulation details at <http://www.chinatax.gov.cn/n810341/n810755/c3453868/content.html> (in Chinese)

⁹ According to the estimates by Zero2IPO, as of the end of August 2022, 1.73% of the GGFs are set up by the central governments, with the remaining at the local government level.

surge of GGFs began in 2015 when the Ministry of Finance (MOF) released the guidelines for the administration of state-owned funds and the State Council decided to set up new GGFs.¹⁰ The statistics from Zero2IPO database show that the average number of new GGFs in each year during the period between 2002 and 2014 is only 27, whilst 314 GGFs were set up in 2015 alone although the growth pace slowed down since 2017. According to Wei et al. (2023), by the end of 2021, there were 1,849 GGFs accounting for 13.2% of the total number of VC funds on the market, with the total target size of CNY 10.18 trillion (US\$1.52 trillion), although a large number of GGFs actually remained ‘inactive’ and made no investments.

Given the crucial role the GGFs play in China’s VC market, it is important to understand the features that distinguish them from other VC funds. First, GGFs mainly rely on government agencies or state-owned investment companies that lack investment experience and competitive incentives to deliver fund management. Analogous to the VC funds common in the equity markets worldwide, a GGF is typically operated in the limited partnership structure where the general partners (GP) make investment decisions and the limited partners (LP) contribute capital and earn returns.¹¹ A GGF’s GP may be a fund management institution established by a government agency, a state-owned investment company, or less often, a third-party professional fund manager.¹² Given the possibility that the governments and poorly incentivized bureaucrats, rather than professional fund managers in practice play a significant role in making GGFs’ investment decisions (Lin, 2021), the management efficiency of GGFs has been questioned.

Second, GGFs have a dual mandate to generate financial returns and to support the government’s policy goals (Luong et al., 2021), which is subject to the criticism of distorting the capital allocation away from projects with the largest growth potential in the economic sense. For instance, to promote the local economic development, a GGF, typically as a fund-of-funds, may require its sub-funds to invest a significant proportion of the total funding in the projects located in the jurisdiction of the government that sponsors the GGF. In return, to attract

¹⁰ See http://www.gov.cn/gongbao/content/2016/content_5051233.htm, <https://www.gov.cn/guowuyuan/gwycwhy201528/> and http://www.gov.cn/guowuyuan/2015-01/14/content_2804136.htm (in Chinese)

¹¹ GGFs can be organized as a Limited Liability Company (LLC), but the proportion of GGFs in LLC form is very low (Chen, 2022).

¹² Chen (2022) documents that about 75% of all GGFs are managed by state-owned firms.

social capital and to incentivize the fund managers to fulfill the requirements of local investments, the GGF may yield part or all of its financial returns earned on the invested projects to the social capital investors and the managers (i.e. the GPs) of the sub-funds. Although the regulations issued in 2015 and 2016 state that GGFs should be operated under market discipline and prohibit governments from intervening daily operations of GGFs, the requirements of local investments are still common in practice (Lin, 2021; Chen, 2022).

Third, GGFs can provide political resources and holistic support for the investee firms. It is recognized that GGFs often obtain and provide local preferential policies, resource linkages, location space and other value-added services for their sub-funds and investee firms through their own connections with relevant government departments (Luong et al., 2021). In other words, GGFs provide supportive resources which are not easily accessed by non-GGF VC funds and may be of critical value to the investee firms.

2.2 China's IPO market

China's IPO market has been characterized by heavy government regulation and there are some qualitative and ambiguous listing criteria for the candidate IPO firms. This may leave room for discretions in the IPO approval process. For instance, IPO candidate firms are required to establish good governance mechanisms and to have the capacity of continuing operation, even for the listings at the Shenzhen ChiNext exchange, the Shanghai STAR exchange, and the Beijing Stock Exchange that have lower bright-line criteria in terms of net profits and firm size, etc. than the traditional stock exchanges.¹³ Under this circumstance, the regulator, the China Securities Regulatory Commission (CSRC), and more specifically, the Stock Issuance Examination and Verification Committee consisting of seven members for each application, of the CSRC plays an essential role in deciding which firms can finally go public (Tian, 2011; Chen et al., 2017) and this makes political connections particularly valuable for firms to obtain IPO application approvals (Liu et al., 2013). The GGFs, with governments at

¹³ See the listing criteria for the listings at the Shanghai Stock Exchange at http://www.sse.com.cn/lawandrules/sselawsrules/stock/main/listing/c/c_20220107_5679269.shtml (in Chinese), for the listings at the Shanghai STAR Exchange at <http://www.csrc.gov.cn/csrc/c106256/c1653997/content.shtml> (in Chinese), and for the listings at the Main Board of Shenzhen Stock Exchange, the ChiNext exchange at <http://www.szse.cn/ipo/guide/requirements/index.html> (in Chinese) and at the Beijing Stock Exchange at http://www.gov.cn/zhengce/zhengceku/2021-10/31/content_5647940.htm (in Chinese).

various levels being their capital providers, have natural connections with governments and may be in an advantageous position in terms of getting IPO approvals or obtaining information regarding IPO applications over the VCs without governmental support.

A stylized fact is the prevailing high initial return (i.e. the stock return on the first day of trading) of IPOs. The initial return of Chinese IPOs is extraordinarily high compared with other main markets. Kimbro (2005), Song et al. (2014) and Qian et al. (2022) document an average initial return on IPOs in China of 132% during 1995-2002, of 66.3% during 2006-2011 and of 172% during 1990-2021.¹⁴ While high IPO initial return is also prevalent in other markets, it is believed that IPO pricing regulations and investor sentiment in particular contribute to the extremely high IPO initial return in China's market (Qian et al., 2022).

At the inception of China's IPO market in the early 1990s, the IPO offer price was determined in an administrative way, by a formula multiplying a P/E ratio of 13 to 15 set by the CSRC and a company's earnings over the past three years, while the pre-set P/E ratio was much lower than the market-wise P/E ratio of traded stocks (Chueng et al., 2009). In 1999, the book building approach, the dominant method of IPO pricing in most countries, was introduced to China. However, this more market-oriented approach was suspended and replaced with the administrative pricing approach since 2001 as the inability of the institutional and the retail investors to deliver reasonable IPO valuations resulted in overheated IPOs with very high offering P/E ratios in 2000. To limit the overvaluation of IPOs, the IPO pricing reform in 2006 modified the book building approach by setting the cap of the offering P/E at 30 (Song et al., 2014) which was subsequently lifted in 2009.¹⁵ The release in the offering P/E ratio cap again led to unreasonably high offering price, and this facilitated the issue of the regulation in 2012 that the offering P/E ratio should not, in general, exceed 125% of the average trading P/E ratio for a given industry. While further modifications in the IPO pricing approach have been made since 2012, the IPO pricing in China is still subject to regulatory intervention (known as 'Window Guidance'), with the maximum offering P/E ratios far below what investors were willing to pay for most IPOs (Qian et al., 2022).

¹⁴ In comparison, for example, the average initial return of IPOs for the US is 17.5% during 1960-2021, 15.7% for the UK during 1959-2020, and 40.5% for Hong Kong during 1980-2021. See the statistics for other countries at <https://site.warrington.ufl.edu/ritter/files/IPOs-International.pdf>.

¹⁵ See the regulation details at <http://www.csrc.gov.cn/csrc/c101802/c1004899/content.shtml> (in Chinese).

While all the IPOs are subject to the same IPO pricing regulations, the investor sentiment may vary across firms, in particular when China's stock market is dominated by retail investors. As at late 2022, the retail investors accounted for about 60% of China's stock market trading, which is over twice the value for the US market (with retail investors accounting for 25% of total stock market trading in 2021) and triple that for the UK market (with retail investors accounting for 20% of total stock market trading in 2020).¹⁶ In this context, how investors perceive the role of GGFs on nurturing the investee firms could affect the initial returns to a large extent, as it is recognized that retail investors are more inclined to make sentiment-driven trades (e.g. Chiang et al., 2010; Neupane et al., 2014). For instance, if the investors value the pros, e.g. political resources and value-added services provided by the GGFs more than the negatives, e.g. the management inefficiency associated with governmental interference, the aggregate optimistic sentiment to the role of GGFs could create value for the investors of GGF-backed IPO firms by driving up initial returns, and vice versa.

3. Literature review and hypothesis development

Venture capital funds sponsored by governments (typically known as governmental venture capital funds, GVC¹⁷), as a unique type of VCs, have been introduced to many countries to foster the growth of entrepreneurial firms and local economy, e.g. the SBIR program in the US, the IIF program in Australia, the LSVCCs in Canada, and the Yozma program in Israel. Meanwhile, numerous studies evaluate the value effects of these programs and mixed results are documented. On the one hand, some evidence suggests that the GVCs spur the sales growth and employment of the investee firms (Lerner, 2000), boost the development of the venture capital industry through increasing the number of new VC-backed startups and VC-backed exits (Avnimelech and Teubal, 2006) and through increasing the total VC finance when GVCs partner with private VCs (Brander et al., 2015), and providing monitoring and value-added

¹⁶ See <https://www.reuters.com/markets/asia/china-retail-investors-gen-z-retirees-sit-out-stock-rally-2023-02-15/>, <https://www.bnymellonwealth.com/insights/the-rise-of-retail-traders.html> and <https://www.sharesoc.org/investor-academy/advanced-topics/uk-stock-market-statistics/>.

¹⁷ Definitions of governmental venture capital funds vary across literature (see a more detailed description in Cumming et al. (2017)). Our paper uses the term in a broad sense to refer to the venture capital funds receiving direct investments from governments regardless of governance forms.

advice to investees (Cumming, 2007). On the other hand, evidence is also found that the GVCs perform worse in spurring the growth of, and achieving better exit performance (assessed by the form of exit, i.e. IPOs, trades or liquidations) for the portfolio firms than private VCs (Grilli and Murtinu, 2014, 2015; Alperovych et al., 2015; Cumming et al., 2017). Despite the extensive evidence documented for the effects of governments exerted on western VC markets, scant attention has been paid specifically to the Chinese market where governments play a central role in capital allocation.

An exception is Colonnelli et al. (2022) that surveys the perception of professional fund managers (GPs) on collaborating with capital providers (LPs) having governmental ties in China's VC and PE markets. They show that the average GP dislikes investors with government ties, indicating that the benefits of political connections for fund managers are small when compared with the negatives of being intervened and controlled by the governments. However, GPs are not the only market participant in the VC market that could be affected by governments. The growth of the investee firms and the wealth of millions of public investors if the firms get listed on stock exchanges may also be affected by the participation of governments in VC investments. To shed some light on this issue, this paper examines the role of Chinese GGFs in value creation for investee firms in the context of initial public offerings.

The pricing of shares at the initial offering determines the proceeds the issuing firms garner from the capital raising, the short-term return the outside investors earn from purchasing the new shares and also the wealth transferred from the shareholders of the issuing firms to public investors (Park et al., 2015). While the issuing firms have strong intention to minimize the 'money left on the table' by minimizing the difference between the closing price on the first day of trading and the offer price, substantial evidence has been found of high initial return prevalent in most of the equity markets in the world (e.g. Beatty and Ritter, 1986; Kimbro, 2005; Song et al., 2014; Qian et al. 2022).

This puzzle of high IPO initial return has been explained mainly in the theoretical frameworks of information asymmetry and investor sentiment. The information asymmetry hypothesis states that IPO firms need to suppress the offer price to induce investors to participate in the share purchases since outside investors tend to value firms conservatively

when they possess less information than the firms (Rock, 1986; Benveniste and Spindt, 1989; Hanley, 1993; Ritter and Welch, 2002; Derrien, 2005; Banerjee et al., 2011; Buchner et al., 2019). Another strand of literature shows that in an environment of high information asymmetry, underpricing could be a signal of firm quality because only high-quality firms are expected to recoup the loss at initial offerings after their performance is realized (e.g. Allen and Faulhaber, 1989; Grinblatt and Hwang, 1989).

However, it is recognized that information asymmetry cannot fully explain the high initial return and that optimistic investor sentiment could drive up the valuation of IPO firms (i.e. the closing price at the first day of trading), contributing largely to the high initial return (Ritter and Welch, 2002; Sanders and Boivie, 2004; Derrien, 2005; Cook et al., 2006; Song et al., 2014). Furthermore, Derrien (2005) relates the extent of underpricing (i.e. the pricing of IPO shares below the true value in the primary market) to investor sentiment by demonstrating that the bullish sentiment at the time of offering makes it less necessary for firms to underprice their shares to attract investors.

The effect of investor sentiment may be in particular pronounced in China. This is because China's stock market is dominated by retail investors who are more likely to make sentiment-driven trades than institutional investors (Chiang et al., 2010; Neupane et al., 2014), exerting particularly strong effects on market overvaluation at IPOs (Derrien, 2005; Ljungqvist et al., 2006; Dorn, 2009; Bajo and Raimando, 2017; Qian et al., 2022). Consistently, Song et al. (2014) find that optimistic investor sentiment plays a dominant role over information asymmetry in explaining the extremely high initial return in the Chinese market, by showing that the average initial return of 66% for the IPOs of 948 Chinese firms can be attributed to an underpricing of 14-22% and an overvaluation of 44-53%.

Given the important role VCs play in the IPO market, there is an ongoing discussion on how VCs affect the initial return. On the one hand, evidence is found that VCs can reduce initial return by restraining IPO underpricing. Under the information asymmetry hypothesis, VC sponsors have the ability to credibly communicate information about a firm's true value, which reduces the information asymmetry and thus IPO underpricing (Neus and Walz, 2005; Tian, 2011). In addition, the signaling hypothesis suggests that the outside investors would perceive

VC sponsoring as a certification of firm quality since only high-quality firms could obtain funding from VCs, and therefore less underpricing is needed to attract investors (Megginson and Weiss, 1991; Park and Patel, 2015). On the other hand, Lerner (1994) and Ball et al. (2011) argue that VCs are proficient in taking their portfolio firms public near market peaks and therefore predict a positive effect of VC sponsoring on market valuation and consequently on initial returns.

Would the injection of GGF funds alter the impact of VC sponsoring on IPO underpricing and overvaluation? We propose that GGFs could make a difference in two different ways. On the one hand, the political connections inherent in GGFs are scarce resources for the growth of investee firms in particular in the Chinese market. Piotroski and Zhang (2014) argue that Chinese politicians tend to give helping hands through, for instance, access to capital, government purchasing, and the authority to examine and approve projects (Tian et al., 2008), to firms in their jurisdictions in order to stimulate local economy and consequently improve their political credentials. Consistently, Wang and Wu (2020) find that firms backed by politically-connected VCs (i.e. VC companies having top executives who serve as country head or higher) have a higher likelihood of obtaining IPO approvals from Chinese Securities Regulatory Commission (CSRC). Also, Cao et al. (2016) document a positive relationship between political connections and the operating performance of entrepreneurial firms in the Chinese market. Thus, if investors recognize the valuable resources GGFs could bring to the investee firms, they may hold more optimistic attitudes towards GGF-backed firms. This, in line with the investor sentiment hypothesis, predicts a stronger valuation-amplifying effect of GGF-backed VC sponsoring than non-GGF-backed VC sponsoring. Consistently, the GGF sponsoring could act as an additional layer of certification of firm quality in comparison with general VC sponsoring without governmental support. This, drawing upon the signaling hypothesis, predicts a stronger underpricing-reduction effect of GGF-backed VC sponsoring than non-GGF-backed VC sponsoring. Following this line of thought, we hypothesize that (i) GGF sponsoring has a positive effect on IPO firms' market overvaluation, (ii) GGF sponsoring has a negative effect on firms' IPO underpricing, or (iii) both occur.

On the other hand, being sponsored by GGF-backed VCs can also be a signal of low firm

quality for two reasons. First, the mandate of GGFs to serve the government's policy goals suggests that GGFs' distribution of capital may not be driven by profit motives and thus could distort the capital allocation away from projects with the greatest growth potential in the economic sense. The lack of economic incentives may also reduce GGFs' motivation to perform the coaching and monitoring activities that are documented to be beneficial for the investee firms (Groh et al., 2010; Dihiya and Ray, 2012; Bernstein et al., 2016; Cummings et al., 2017). Second, most of the GGFs are managed by state-owned firms or governmental agencies (as discussed in Section 2) which are believed to be more likely to make inefficient investment decisions. Even if GGFs cooperate with professional fund managers, anecdotal evidence shows that there is deep governmental intervention in daily fund management. Thus, such negative signals for firm quality may spur the conservative attitudes of investors towards IPOs backed by GGFs, leading to lower IPO overvaluation. Consistently, the public investors may perceive the GGF sponsoring as a signal of increased uncertainty about firm quality, making it more necessary for GGF-backed firms to depress the offer price in order to attract investors at IPOs, resulting in higher IPO underpricing. Following this line of thought, we hypothesize that (i) GGF sponsoring has a negative effect on IPO firms' market overvaluation, (ii) GGF sponsoring has a positive effect on firms' IPO underpricing, or (iii) both occur.

In summary, GGFs could exert impact on IPO overvaluation (underpricing) in two opposite ways, hence it is an empirical question of which force dominates in the practice. An equally important, if not more, question is whether the impact of GGFs on the stock performance at IPOs persists for a long period post IPOs. Moreover, besides exerting impacts on firms' performance in the financial market, do GGFs also affect firm operating performance and innovation activities in the product market? These questions will be examined in details in the empirical analysis section.

4. Data collection

The VC investment data covering the period between 2010 and 2021 were collected from ZeroIPO, which is documented to provide the most comprehensive and accurate data for

China's VC market (Chen, 2022; Colonnelli et al., 2022).¹⁸ For each VC investment, the information was collected for the name of the investee firm and of the VC company, the date and the CNY amount of the investment, the industry the investee firm comes from, the province the investee firm is registered in, and whether a given investment is funded by any GGFs (referred to as GGF-backed investment hereafter). In total, the data for 23,139 investments to 9,059 firms made by 2,704 VC companies were collected, of which 3,585 investments to 2,494 firms made by 376 VC companies were funded by GGFs. Table 1 reports the descriptive statistics of the VC investments by year (Panel A) and by industry (Panel B), for all the investments (left panel) and for the GGF-backed investments (right panel) respectively.

Panel A shows that the number of the VC investments (investee firms) surged over the past decade, from 141 (74) in 2010 to 7,165 (3,412) in 2021 at an average annual growth rate of 46.7% (44.9%), and the amount of the investments grew by 7,500% from 1.52 billion CNY in 2010 to 116.64 billion CNY in 2021. Such a rapid growth is also observed and even more prominent for the GGF-backed investments. The average annual growth rate of the number of GGF-backed investments (GGF-backed investee firms) was 60% (56%), and the CNY amount of the GGF-backed investments increased by over 250 times over the 12 sample years, reaching 12.55 billion CNY in 2021 accounting for 10.8% of all the VC investments on the market. Panel B lists the ten industries that received the most VC investments. It shows that, the (GGF-backed) VC investments centered on the high-technology sectors for the Chinese market, among which the biotechnology and health industries attracted the most (GGF-backed) VC investments, accounting for a quarter of the total investments. Overall, the statistics show the increasing importance of the VCs, in particular the GGF-backed VCs, in nurturing high-tech companies, confirming the significance of understanding whether and how the GGFs could create values for the investee firms.

We obtained the list of IPOs between 2010 and 2021 from China Stock Market and Accounting Research (CSMAR) database. For each IPO firm, the information was collected for the firm name, the initial return, the offering price, the IPO and the listing dates, the stock exchange the firm was listed on, the firm's inception date, total assets and earnings per shares

¹⁸ Chen (2022) and Colonnelli et al. (2022) provide detailed descriptions of the Zero2IPO database.

as stated in the prospectus, whether the firm is state-owned or not, the underwriter(s) of IPOs, and the firms' yearly return on assets, return on equity, the ratio of earnings before interest and taxes (EBIT) to total assets, the ratio of R&D expenditure to total revenue, and the number of patents applied for and obtained during the post-IPO period . To identify whether an IPO firm was funded by VCs and if so, the list of the VC investments prior to the IPO, the Zero2IPO data and the CSMAR data were merged by firm names. The last columns of the two panels in Table 1 report the proportion of the VC-backed firms that went public as of 2021. It shows that on average, 6.9% of all the VC-backed firms went public successfully whereas the proportion was 10% for the GGF-backed firms. Such a higher probability of IPO success for the GGF-backed investee firms was exhibited in almost every single year and every industry (except for the investments made in years 2010 and 2021, and in the machinery manufacturing and financial service industries). These descriptive statistics provide supportive evidence for the conjecture that the support from GGFs could bring about valuable political resources for firms which may increase the probability of obtaining approvals of IPOs on the Chinese market.¹⁹

We excluded financial firms and those with missing data after the merge of the Zero2IPO with the CSMAR databases, leaving 2,855 distinct IPO firms. The first column of Table 2 reports the total number of the IPO firms by year during the sample period of 2010-2021, and the second and the third column shows the number of IPO firms that were funded by VCs, and funded by GGFs, respectively.²⁰ Out of the 2,855 sample firms during the sample period, 16% were funded by VCs and 6% were funded by GGFs prior to the IPOs.

*****Insert Table 1 here*****

*****Insert Table 2 here*****

¹⁹ The impact of being sponsored by GGFs on the probability of obtaining IPO approvals is out of the scope of this paper. Yet, to test the relationship more formally, Table A.1 in the Appendix reports the results for the regressions of the probability of achieving IPOs on whether a firm receives funding from GGFs. and after controlling for the characteristics of the VC investments and VC companies (due to the data unavailability of the characteristics of unlisted firms, firm characteristics are not controlled for). The coefficients of the GGF dummy are positive and statistically significant in both specifications, suggesting that the support from the GGFs increased the probability of firms getting IPO approvals.

²⁰ No firms went public in 2013 when the IPO approval was suspended by CSRC.

5. Variable construction and methodology

5.1 Variables and summary statistics

The dependent variable, the initial return of an IPO ($Return_{int}$) is calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price, consistent with the common practice in the existing literature (e.g. Tian, 2011; Tian, 2012; Song et al., 2014; Lehnertz et al., 2022). To assess whether the initial return is driven by ‘true’ underpricing relative to a firm’s fair value or by market overvaluation, $Return_{int}$ is decomposed into the IPO underpricing ($Underprc$), i.e. the difference between a firm’s intrinsic value and offer price divided by the offer price, and the IPO overvaluation ($Overval$), i.e. the difference between the first-day closing price and the intrinsic value divided by the offer price. Consistent with the practices adopted in the existing literature (e.g. Purnanandam and Swaminathan, 2004; Song et al., 2014), a firm’s intrinsic value is estimated by Eq. (1):

$$IV_i = EPS_i * (P/E)_{i_indmedian}, \quad \text{Eq. (1)}$$

where EPS_i is firm i ’s earnings per share as stated in the prospectus, and $(P/E)_{i_indmedian}$ is the median price-to-earnings ratio across the firms of the same industry as firm i traded on the market in the month when the IPO of firm i took place.

The independent variables include the dummy indicating an IPO firm was funded by any VC companies prior to the IPO (VC), and the dummy indicating an IPO firm was funded by any GGFs prior to the IPO (GGF). To control for the firm, underwriter, and listing characteristics that are documented to affect initial returns (e.g. Chan et al., 2004; Tian, 2011; Tian, 2012; Song et al., 2014; Chen et al., 2017; Buchner et al., 2019; Lehnertz et al., 2022), the following variables are constructed. A firm’s age (Age) is defined as the number of years it has been in operation till the IPO date. A firm’s total assets ($Asset_{preIPO}$) and earnings per share (EPS_{preIPO}) prior to the IPO is the value of total assets and the value of the EPS disclosed in the prospectus, respectively. A dummy SOE indicates that an IPO firm is a state-owned firm. $Daygap$ is defined as the number of days between the offering and the listing dates, given that it is found that the longer time gap between the two dates, the higher the initial return is, as

investors require a higher return for the larger risk they bear during the gap period (Chan et al., 2004; Chen et al., 2017). The ability and reputation of the lead underwriter of a given IPO is proxied by the sum of the CNY amount of IPOs for which it had been the lead underwriter prior to the current IPO ($Value_{udw}$). To consider the effect of the market sentiment, a dummy $IPOhot$ is defined as the ‘hot’ IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. For the VC-backed IPOs, the characteristics of the VC investors are also controlled for. Age_{VC} is defined as the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. $\#Fund_{VC}$ is defined as the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO.

Table 3 Panel A reports the summary statistics of the variables defined above for the VC-backed IPO firms and the non-VC-backed IPO firms, respectively, and Panel B further classifies the VC-backed firms into two sub-groups of the GGF-backed firms and non-GGF-backed firms and reports the analogous statistics. Panel A shows that the VC-backed firms are older, more profitable pre-IPO, less likely to be SOE, and tend to be underwritten by more reputable underwriters and to be listed during hotter period than the non-VC-backed firms. Moreover, the average initial return of the VC-backed IPOs is over twice that of the non-VC-backed IPOs (146.1% versus 59.8%), and it is driven by the larger market overvaluation rather than higher underpricing of the VC-backed IPOs. To be specific, the *Overval* (*Underpr*) of the average VC-backed IPO is 159.8 (73.6) percentage points larger (smaller) than the average non-VC-backed IPO, and the differences are both statistically significant at 1 percent.

*****Insert Table 3 here*****

At a more granular level, Table 3 Panel B reports the analogous statistics to those as shown in Panel A for the GGF-backed IPO firms and non-GGF VC-backed IPO firms separately. It shows no statistically significant differences in the firm characteristics or the underwriter characteristics between the two groups of firms. Yet, there is some evidence that the GGF-backed firms are more likely to receive investments from larger VC companies (indicated by

$\#Fund_{VC}$). Importantly, the initial return of the average GGF-backed IPO is 45.2 percentage points larger than that of the average non-GGF VC-backed IPO firm, which is driven by the higher market overvaluation (130.8% for the GGF-backed IPOs and 54.7% for the non-GGF VC-backed IPOs) rather than underpricing (43.6 % for the GGF-backed IPOs and 74.5% for the non-GGF VC-backed IPOs) for the GGF-backed IPOs. The evidence implies that the market values the political resources and value-added services provided by the GGFs more than the possible capital distortion and management inefficiency they bring. Consistently, the lower underpricing of the GGF-backed IPOs suggests that less underpricing is needed to achieve successful capital raising since the political resources underlying the GGF investments are already a device to increase the probability of successful IPOs.

5.2 Methodology

To mitigate the endogeneity issue associated with the possibility that covariates affect both the probability of a firm being invested by VC funds and the initial return at IPOs, we perform the entropy balancing matching (EBM) following recent literature (e.g. McMullin and Schonberger, 2020; Beck et al., 2022; Basu et al., 2022). The recently developed methodology enables to eliminate the inequalities in representation with respect to the first, second or higher moments of the covariate distributions through reweighting the sample units (Hainmueller, 2012; Hainmueller and Xu, 2013). In addition, it is recognized as a more effective matching technique than conventional propensity-score matching and nearest-neighbor matching etc. in terms of achieving covariate balance and retaining more information (e.g. Hainmueller, 2012; Cazier et al., 2020). In the empirical analysis where EBM is applied, following the existing literature, we match the treatment group and the control group based on all observable characteristics (i.e. the control variables as defined in Section 5.1).

6. Do government guidance funds create value in the financial market?

6.1 Effect on IPO initial return

6.1.1 Baseline results

We start with testing the effect of having VC investments on the initial return of IPOs to shed some light on the value effect of the VC investing at large on China's IPO market. To be specific, OLS regressions of $Return_{int}$ on the VC dummy are adopted after controlling for various firm characteristics (i.e. Age , $Asset_{preIPO}$, EPS_{preIPO} , SOE), listing characteristics (i.e. $Daygap$), underwriter characteristics (i.e. $\ln(Value_{udw})$) and market sentiment variables (i.e. $IPOhot$) as defined in Section 5.1, plus the dummies indicating IPO years, the industries firms belong to and listing exchanges. Then, in order to assess whether the relationship between VC investments and initial return is driven by underpricing or market overvaluation, we replace the dependent variable $Return_{int}$ with the variable $Underprc$ and $Overval$, respectively. Table 4 Panel A reports the results for the three specifications, with the column name indicating the dependent variable of question.

The first column of Panel A shows positive and weakly statistically significant (at 10 percent level) coefficient for the VC dummy, indicating that being invested by VC funds prior to IPOs increases the initial return of IPOs by 16.9%. Importantly, the coefficient of the VC dummy is negative (positive) and statistically significant at 1 percent in the specification for $Underprc$ ($Overval$), suggesting that it is the market overvaluation rather than the underpricing that drives the positive effect of VC sponsoring on the IPO initial return. There are several not mutually exclusive explanations for the evidence observed. First, drawing on the information asymmetry hypothesis (e.g. Beatty and Ritter, 1986; Loughran and Ritter, 2004), the negative relationship between VC investments and underpricing indicates that as professional financial intermediaries, VC sponsors have the ability to credibly communicate information about a firm's true value, which reduces the information asymmetry and thus IPO underpricing (Neus and Walz, 2005; Tian, 2011). Second, in line with the signaling hypothesis (e.g., Spence, 1973; Allen and Faulhaber, 1989; Connelly et al., 2011; Park and Patel, 2015), VC sponsoring could

be a certification of firm quality, which is documented to reduce IPO underpricing (Megginson and Weiss, 1991). Third, under the investor sentiment hypothesis, the positive effect of VC investments on overvaluation is in line with the argument that VC sponsors are proficient in taking their portfolio firms near market peaks (Lerner, 1994; Ball et al., 2011), and/or that investors hold optimistic opinion on firms' growth potential when the firms could obtain funding from VCs.

The results for the control variables are consistent with the existing literature. Specifically, there is a positive relationship between the time from offering to listing (indicated by the positive coefficient of *Dapgap*, statistically significant at 1 percent) and IPO initial return, suggesting that the investors require a higher return for the larger risk they bear during the gap period (Chan et al., 2004; Chen et al., 2017). The positive coefficients of $\ln(Value_{udw})$ in the specifications for *Return_{int}* and for *Overval* indicate that underwriters' reputation has a positive impact on market overvaluation and consequently on initial returns. The evidence is consistent with the argument that investors perceive good underwriter reputation as a positive signal of firm quality (Song et al., 2014), which may spur investors' optimistic sentiment towards the IPO firms underwritten by intermediaries with good reputation. The positive and statistically significant (at 1 percent) coefficients of *IPOhot* in the specifications for *Return_{int}* and for *Overval* are consistent with the literature and intuition that being listed during 'hot' period could increase market overvaluation and thus IPO initial returns. In addition, consistent with the findings in Song et al. (2014), we also document that IPO initial returns are negatively associated with firm assets (*AssetpreIPO*) and firm EPS (*EPSpreIPO*) prior to IPOs.

More importantly, to test whether and how the GGF capital alters the effect of VC sponsoring, we repeat the regressions analogous to those in Table 4 Panel A but on the subsample consisting of the firms that had GGF-backed VC investments plus those that did not have VC investments prior to IPOs, and on the subsample consisting of the firms that had non-GGF-backed VC investments plus those without VC investments prior to IPOs separately. Table 4 Panels B and C report the results for the two subsamples, respectively.

Panels B and C show distinct differences in the coefficients of the *VC* dummy in both statistical and economic sense. To be specific, the *VC* coefficient in the specification of *Return_{int}*

is statistically significant in Panel B only, and its magnitude (0.368) is nearly six times that (0.063) as shown in Panel C. The evidence indicates that the observed positive relationship between VC sponsoring and initial return at IPOs is driven by the VC investments with GGF capital only. Furthermore, Panels B and C show that the absolute value of the VC coefficient in Panel B is larger than that reported in Panel C for the specifications of both *Underprc* (-0.418 in Panel B versus -0.190 in Panel C) and *Overval* (0.804 in Panel B versus 0.268 in Panel C), with the differences being highly statistically significant, indicated by the seemingly unrelated estimation (SUE) results.

The evidence supports the conjecture that investors hold optimistic attitudes towards GGF sponsoring, indicating that the market values the benefits of political connections more than the costs of management inefficiency and capital misallocation underlying the GGF investments when evaluating new shares. While receiving GGF capital strengthens both the negative effect of VC sponsoring on underpricing and the positive effect on overvaluation, the force for the overvaluation dominates that for the underpricing, leading to net positive effect of GGF-backed VC sponsoring on initial return.

*****Insert Table 4 here*****

*****Insert Table 5 here*****

To address the endogeneity concern associated with the possibility that the factors affecting the probability of a firm being sponsored by VCs simultaneously affect the firm's initial return of IPO, we repeat the regressions analogous to those in Table 4 but using weights obtained from the entropy balance matching on all the independent variables except for the *VC* dummy. Table 5 reports the results. The results for the control variables remain similar to those presented in Table 4 although the effects of *Age*, *Asset_{preIPO}*, *EPS_{preIPO}* and *IPOhot* become stronger in some specifications. Importantly, the results for the key variable of interest (the *VC* dummy) observed in Table 4 still preserve after the EBM, confirming the robustness of the results.

6.1.2 Robustness checks

We employ a variety of the Heckman (1979) two-step estimation – the treatment effect

model, as an alternative approach to mitigate the endogeneity issue associated with self-selection bias. In the first-step model, probit regression is adopted to estimate the likelihood a firm being backed by a VC. In the second-step model, we include the inverse Mills ratio (*IMR*) obtained from the first-step estimation as the additional independent variable in the specifications analogous to those in Table 4. To achieve the first-step estimation, we construct an instrumental variable *#firm_prov* indicating the number of firms invested by VCs in the province a given IPO firm is registered.²¹ This instrumental variable is chosen because location factor such as the economic development, business environment and transportation convenience etc. is crucial for VCs to make investment decisions (e.g. Cumming and Dai, 2011; Wang and Wu, 2020). The variable *#firm_prov* could indicate the tendency of a VC making investments in the province of a firm of question, and thus correlates with the probability of the firm being backed by VCs. Moreover, this variable is unlikely to affect initial returns at the firm level except via VC backing.

Table 6 reports the results of the Heckman two-step regression. In each panel, the first column shows the results for the first-step estimation and the last three columns report the results for the second-step estimation. It shows that the firms being backed by VCs are more likely to be located in provinces with more VC investment activities (indicated by the three positive coefficients of $\ln(\#firm_prov)$, two of them statistically significant at 1 or 5 percent), making it a valid instrument variable for the Heckman (1979) adjustment procedure.²² Importantly, the second-step regressions show that the results for the key variable of interest (*VC dummy*) still preserve after adding the *IMR* obtained from the first-step estimation. Hence, the main finding of a positive relationship between VC sponsoring and initial return which is in particular pronounced for GGF-backed VC sponsoring still preserves after correcting possible selection bias.

*****Insert Table 6 here*****

As another robustness check for the baseline results, we use an alternative method to

²¹ When the subsample of (non-) GGF-backed VC investments is assessed, *#firm_prov* is defined as the number of firms invested by (non-) GGF-backed VCs in the province a given IPO firm is registered correspondingly. The natural logarithm of *#firm_prov*, i.e. $\ln(\#firm_prov)$ is used in the regressions.

²² The variable $\ln(\#firm_prov)$ is not statistically significant if it is added to the baseline regressions analogous to the specifications in Table 4 (untabulated to save space), indicating that it does not correlate with the dependent variable in the second-step model.

estimate firms' intrinsic values, to mitigate the concern that the main findings are sensitive to the estimation of firms' intrinsic values. To be specific, we estimate a firm's intrinsic value using the mean P/E ratio across the firms of the same industry it comes from when applying Eq. (1), rather than the median P/E ratio as used in the main analysis, and repeat the regressions analogous to Tables 4 and 5. To save space, the results are reported in the Appendix Tables A.2 and A.3, respectively. The two tables show that regardless of whether the entropy balancing matching is performed or not, the finding still preserves that GGF sponsoring has a stronger positive (negative) effect on market overvaluation (IPO underpricing) than other-VC sponsoring.

6.1.3 Additional evidence

To directly compare the effect of GGF-backed and non-GGF-backed VC sponsoring on initial returns, we repeat the regressions on the subsample of IPO firms that receive VC investments prior to IPOs, replacing the *VC* dummy with the *GGF* dummy indicating a firm having GGF-backed VC investments pre-IPO. In addition, as previously mentioned, since the VC companies that invest in GGF-backed and non-GGF VC-backed firms differ in their operation history and management experience, we add the variables *Age_{VC}* and *#Fund_{VC}* as defined in Section 5.1 to control for VC company characteristics. This could rule out the possibility that the results are driven by characteristics of VC companies rather than investors' perceptions on GGF sponsoring. To ensure robustness and save space, we report only the results after performing the EBM on all the independent variables except for the *GGF* dummy. Table 7 Panels A and B report the results without and with *Age_{VC}* and *#Fund_{VC}* in the regressions, respectively.

Panel A shows that the coefficient of *GGF* dummy is positive and statistically significant at 10 percent, suggesting that the initial return of the firms backed by GGFs is on average larger than that of the firms backed by non-GGF-backed VCs by 16.1 percentage points. It also shows that the difference is driven by the higher overvaluation of GGF-backed firms in comparison with the firms backed by non-GGF-backed VCs (indicated by the positive coefficient of *GGF* in the specification of *Overval*, statistically significant at 1 percent). In comparison, the

underpricing-reduction effect of GGF sponsoring is not as strong as its valuation-amplifying effect (the negative *GGF* coefficients in the *Underprc* specifications are statistically insignificant). Similar results remain after controlling for the characteristics of VC companies as shown in Panel B, suggesting that even after keeping the reputation and ability of VC companies equal, the firms sponsored by GGFs still achieve higher market valuation. Hence, the evidence reinforces the finding that investors hold optimistic perceptions on the firms backed by GGFs, indicating that they value the political resources underlying the GGF investments more than the concern of management inefficiency and capital misallocation brought by GGFs.

*****Insert Table 7 here*****

Overall, this section shows a positive effect of the VC sponsoring on the IPO initial return. Specifically, VC investments create value in the financial market for both the IPO firms by increasing the proceed per share they could garner from IPOs and for the investors by increasing the market valuation. More importantly, we find that GGF-backed VC investments, in possession of political resources, could create larger value through spurring market optimistic sentiment to a larger extent than other VC investments.

6.2 Effect by GGF investment stage and investment amount

As discussed in Section 2, GGFs bring about not only the benefits of political resources, but also costs associated with capital misallocation and management inefficiency. In order to examine whether investors' optimistic sentiment towards GGF-backed firms is diluted by the concern for the potential costs associated with the governmental intervention, we classify the GGF-backed firms into subgroups according to the stage and the amount GGFs invest in the firms and repeat the analysis analogous to that in Table 7 Panel B. The reasons why we look into the stage and amount GGFs invest are as follows.

First, given that GGFs have the intention to support governments' policy goals and are not so incentivized to pursue profits as other VC funds, if a firm's early VC investments are made by GGFs rather than market-oriented VCs, the market may regard it as an uncompetitive firm that lacks attractiveness to, and needs support from GGFs as an additional 'selling point' to

attract attention from profit-seeking VCs. Following this line of thought, we perform analysis on the firms that received GGF investments as their first VC investments (denoted as ‘GGF-first’ group) and on the firms that received GGF investments at a later stage (denoted as ‘GGF-late’ group) separately.²³

Second, it is natural to conjecture that the more GGFs invest in a firm relative to other VCs, the more governmental intervention would exist in managing the firm. If the market regards governmental intervention as a signal of management inefficiency, we should observe that the valuation for the IPO firms decreases with the share ownership of GGFs. To assess whether this is the case, we perform the analysis on the firms for which the GGF investments account for more than 50% (denoted as ‘GGF-high’ group), and for less than or equal to 50% (denoted as ‘GGF-low’ group) of the total VC investments a firm received prior to IPOs separately.²⁴

Tables 8 and 9 report the results for the heterogeneity tests by the GGF investment stage and amount after the EBM on all the firm, underwriter listing and VC company characteristics, respectively. Table 8 Panels A and B reports the results for the regressions analogous to those in Table 7 Panel B when the subsample of GGF-first firms and of GGF-late firms is examined, respectively. None of the coefficients for the *GGF* dummy is statistically significant as shown in Panel A, while the higher market overvaluation and consequently the larger initial return of GGF-backed firms relative to other VC-backed firms are observed in the group of the GGF-late firms only (indicated by the coefficient of 0.362 and 0.447 in the *Return_{int}* and *Overval* specifications, statistically significant at 1 and 10 percent, respectively). The finding confirms our conjecture that the market views the GGF investment at an early stage as a negative signal for a firm’s quality and growth potential, weakening the positive effect of GGF sponsoring on creating value for investors.

Table 9 also shows striking differences in the role of GGF sponsoring between the groups of GGF-high and GGF-low firms. To be specific, the statistically significantly positive effect of the GGF sponsoring on *Overval* and *Return_{int}* is observed for the GGF-low group of firms

²³ Summary statistics show that 40 % of the GGF-back IPO firms received GGF investments as their first VC investments, and 76% of them received GGF investments as one of the earliest three VC investments. The statistics suggest that GGFs act as a ‘leading’ role in financing a significant portion of firms, confirming their importance in providing capital to under-funded firms.

²⁴ Summary statistics show that the GGF investments account for 43.9% (35.7%) of the total VC investments a firm received for the average (median) sample firm, and that the GGF investments account for over 50% of the total VC investments for 32.6% of the sample firms.

only. In contrast, the positive effect of GGF sponsoring on *Overval* and *Return_{int}* turns negative for the GGF-high group of firms (statistically insignificant and significant at 1 percent, respectively). It provides support for the conjecture that while the investors value the political resources brought by GGFs, they are also concerned with the deep intervention from governments in managing the invested firms.

Overall, this section shows evidence that the effect of GGF sponsoring on the investor sentiment and thus firms' initial return at IPOs varies with the stage GGFs make investments and with the proportion of shares held by GGFs relative to other types of VC funds in a given firm.

*****Insert Table 8 here*****

*****Insert Table 9 here*****

6.3 Long-run post-IPO stock returns

While the previous sections show that GGF-backed firms enjoy a higher market valuation at IPOs, does it arise from a temporary overvaluation by the market or from real benefits GGFs could bring up to effectively improve firms' long-term performance? If the political resources and value-added services provided by GGFs do not support the long-term development of the invested firms, one would expect the disappearance of, or even reversal of the outperformance of GGF-backed firms at IPOs in the long-run. To shed some light on the dynamics of the post-IPO stock performance of the sample firms in different groups, we first calculate the monthly stock return of a given sample firm in excess of the median return of the stocks in the same industry the firm of question comes from, before taking the average abnormal return across all the firms in a given group (i.e. GGF-backed firms, non-GGF VC-backed firms and non-VC-backed firms). Figure 1 plots the average abnormal monthly return for each group of firms in the 24 months post IPOs.

It shows a similar pattern for the three groups of firms that the abnormal return of the newly offered stocks declined over time, as an indication of the cooling-down of the optimistic sentiment with the arrival of information about the true value of the firms post IPOs, consistent with the existing literature (e.g. Ritter, 1991; Levis, 1993; Chahine, 2004, 2008). On average,

the firms of the non-VC-backed group underperform the other two groups in most of the two-year period, confirming the ability of VCs to improve long-term firm performance documented in the literature (e.g. Brav and Gompers, 1997; Ivanov and Xie, 2010; Tian, 2011; Chemmanur et al., 2014). Interestingly, although Section 5 shows strong evidence of the outperformance of the GGF-backed firms in initial returns than the non-GGF VC-backed firms, Figure 1 shows that the average abnormal return of the GGF-backed group is higher than that of the non-GGF VC-backed group in only 7 out of the 24 months post-IPO, and the outperformance of the GGF-backed group reverses completely since one year post IPOs. Moreover, after 17 months post IPOs, the GGF-backed firms not only underperform non-GGF VC-backed firms, but also underperform non-VC-backed firms, by 0.025% - 0.161% per month.

*****Insert Figure 1 here*****

Figure 1 already shows preliminary evidence that the high valuation of GGF-backed IPOs does not persist in the long run after IPOs. To formally test the post-IPO long-term stock performance of different groups of firms, we run OLS regressions of the average abnormal return (relative to the median of the industry) over the 24 months post IPOs on the *VC* dummy, after controlling for firm characteristics. Analogous to the method used in Table 5, the analysis is performed on (i) the subsample of GGF-backed firms plus the non-VC-backed firms, and on (ii) the subsample of non-GGF VC-backed firms plus the non-VC-backed firms, using the weights obtained from EBM on all the covariates. Table 10 Panel A reports the results for the two subsamples separately. For the key variable of interest, while the coefficients of *VC* dummy are positive in both specifications, it is statistically significant at 5 percent in specification (ii) only, suggesting that only the non-GGF VC-backed firms outperform non-VC-backed firms in the long-term post IPOs (by 0.5% per month). Moreover, the magnitude of the VC coefficient in specification (ii) is over twice that in specification (i), confirming the outperformance of the non-GGF VC-backed firms over the GGF-backed firms post IPOs.

Therefore, the results suggest that the high valuation by the market towards GGF-backed firms is more of a temporary overvaluation than of a reflection of long-term outperformance of these firms. To provide more granular evidence, we restrict the sample to the GGF-backed firms and examine whether the long-term post-IPO performance of these firms vary across the

stage and amount GGFs invest, with the results reported in Table 10 Panel B. The coefficient of the dummy *GGF_first* is negative and statistically significant at 5 percent, suggesting that the average abnormal return over the 2 years post IPO of the firms to which GGFs make the first VC investment (i.e. GGF-first firms) is lower than that of the firms receiving GGF investments at later stage (i.e. GGF-late firms) by 1.9% per month. This evidence coincides with the observation from Table 8 that investors do not value GGF-first firms as highly as GGF-late firms, justifying investors' concern about the policy motive (i.e. lack of profit motive) underlying the early-stage GGF investments and the associated capital misallocation away from firms with greatest growth potential. Moreover, the coefficient of the dummy *GGF_high* is negative, suggesting that the proportion of GGF investments is negatively associated with the long-term post-IPO stock performance of GGF-backed firms, albeit statistically insignificant. The evidence justifies investors' concern about the heavier governmental intervention associated with more GGF investments in managing the invested firms, reflected by the lower valuation of these firms at IPOs in comparison with firms with less GGF investments (as indicated by Table 9).

*****Insert Table 10 here*****

Overall, this section shows that the high market valuation of GGF-backed firms is more likely to arise from a temporary overvaluation by the market rather than from the firms' persistent growth potential. The outperformance of stocks at IPOs of GGF-backed firms over non-GGF VC-backed firms and even over non-VC-backed firms disappears within two years following IPOs. The evidence implies that while investors put more weight on the benefits of political resources brought by GGFs than the costs of governmental intervention when evaluating the role of GGF sponsoring at IPOs, the additional resources brought by GGFs fail to improve the long-term performance of their invested firms in the financial market.

7. Do government guidance funds create value in the product market?

7.1 Post-IPO operating performance

This section investigates whether GGF sponsoring creates value for firms in the product

market by comparing the operating performance of different groups of firms following IPOs. To do so, for each firm, we calculate the average annual operating performance, i.e. return on asset (*ROA*), return on equity (*ROE*) and the ratio of EBIT to total assets (*EBIT/assets*) within two years following IPOs. Then, we run OLS regressions of the operating performance on the *VC* dummy controlling for firm characteristics when the GGF-backed firms and non-GGF VC-backed firms are examined separately, using the weights obtained from EBM on all the covariates. The results for the two subsamples are reported in Table 11 Panels A and B, respectively. All the three measures for the operating performance are used as the dependent variables to test the robustness of results.

Panels A and B show that the IPO firms with longer operational history, smaller size and higher earnings per share prior to IPOs have better operating performance post IPOs. Importantly, the two panels show striking differences in the coefficients of the *VC* dummy. All of the coefficients of the *VC* dummy are positive in Panel B, albeit statistically insignificant, but they are negative and statistically significant at 1 or 10 percent in all the specifications of Panel A. Specifically, the average ROA of the GGF-backed firms in the two years following the IPOs is lower than that of the non-VC-backed firms by 0.6% per annum (the value for ROE and EBIT is 1.6% and 0.9%, respectively). The evidence suggests that VC sponsors do not create additional product market value for their invested firms, and even worse, the GGF-backed VCs erode the operating performance of their invested firms significantly in both statistical and economic sense.

To show additional evidence, we restrict the sample to the VC-backed firms and compare the operating performance between the GGF-backed firms and non-GGF VC-backed firms, after controlling for firm and VC company characteristics, using the weights obtained from the EBM on all the covariates. The results are reported in Table 11 Panel C. It shows that the longer history of operating VC business does not translate to better support for the invested firms to improve their operating performance (indicated by the negative coefficients of *Age_{VC}*). Importantly, the coefficients of the *GGF* dummy are all negative (statistically significant at 1 or 5 percent) regardless of the variable used to measure operating performance. It suggests that the GGF-backed firms underperform the non-GGF VC-backed firms by 1.8% (2.7% or 2.4%)

per annum when *ROA* (*ROE* or *EBIT/asset*) is assessed. The underperformance of GGF-backed firms in the product market is in line with their underperformance in the stock market post IPOs in comparison with the firms backed by non-GGF-VCs indicated by the results in Table 10.

Overall, this section adds support to the conclusion drawn in the last section that GGF-backed VCs fail to take advantage of the political resources and value-added services provided by governments to spur the long-term growth of their investee firms. Rather, the governmental intervention and management inefficiency coming along with the GGF capital dominate the benefits, if any, being detrimental to firm growth.

*****Insert Table 11 here*****

7.2 Effect by GGF investment stage and investment amount

To examine whether the product market value created by the GGF-backed VCs varies across the stage and amount of GGF investments, we restrict the sample to only the GGF-backed firms and assess the differences in the operating performance between the GGF-first and GGF-late firms, and between the GGF-high and GGF-low firms, with the results reported in Table 12 Panels A and B, respectively.

Consistent with the results shown in Table 11 Panel C, Table 12 shows that firms with shorter operational history, larger size, and lower earnings per share prior to IPOs and that are state-owned have worse operational performance post the IPOs. Importantly, Table 12 Panel A shows that all of the three coefficients of the *GGF_first* dummy are negative and statistically significant at 5 or 10 percent, suggesting that the average *ROA* (*ROE* or *EBIT/asset*) over the 2 years post IPO of the firms to which GGFs make the first VC investment is lower than that of the firms receiving GGF investments at later stage by 2.5% (4.2% or 2.8%) per annum. Similarly, Panel B shows that all the coefficients of the *GGF_high* dummy are negative, one statistically significant at 10 percent, suggesting that the proportion of GGF investments is negatively associated with the long-term post-IPO operating performance of GGF-backed firms.

The underperformance of the GGF-first firms and of GGF-high firms in the product market is consistent with their underperformance in the financial market as suggested by the evidence

in Table 10 Panel B. Therefore, this section once more provides evidence giving grounds to the heterogeneity of investors' valuation towards firms at IPO that receive GGF investments at different stages and in different degrees. It also provides additional support for the conclusion that the costs of receiving GGF investments for firm growth vary across when and how much GGFs step in the firms.

*****Insert Table 12 here*****

7.3 Innovation activities

While GGFs underperform in terms of improving firms' operating performance, do they stimulate firms' innovation activities? This question is of particular relevance given the fact that Chinese regulators have stated clear objectives to promote indigenous innovations since years ago.²⁵ Moreover, if the funds financed through GGFs are invested in innovation activities, it is not necessary for the long-term economic benefits associated with innovation activities, if any, to surface in a relatively short period. In other words, it is possible that GGFs, with the intention to support governments' innovation-policy goals, outperform other VC funds in terms of stimulating firms' innovation activities although they underperform in terms of improving firms' operating performance within two years post IPOs. To test whether it is the case, we repeat the regressions analogous to those in Table 11 Panel C but the dependent variable is replaced with measures for firms' innovation expenditure and productivity. To be specific, a firm's innovation expenditure is measured by the average yearly ratio of R&D expenditure to total revenue within two years post IPOs (*RD/revenue*), and its innovation productivity is measured by two variables – the average number of patents applied for per year within two years post IPOs (*Aplpt*), and the average number of patents obtained per year within two years post IPOs (*Obtpt*). The natural logarithm of one plus *Aplpt*, and of one plus *Obtpt* is used in the regressions.

Specifications (i)-(iii) of Table 13 Panel A reports the results when *RD/revenue*, $\ln(1 + Aplpt)$ and $\ln(1 + Obtpt)$ is used as the dependent variable, respectively. It shows that none of

²⁵ See, for instance, the 12th Five Year Plan at https://www.gov.cn/wszb/zhibo444/content_1844173_4.htm (in Chinese), the 13th Five Year Plan at https://www.gov.cn/xinwen/2016-03/17/content_5054992.htm (in Chinese) and the 14th Five Year Plan at https://www.gov.cn/xinwen/2021-03/13/content_5592681.htm (in Chinese).

the coefficients of the main variable of interest - *GGF* dummy is statistically significant, suggesting that GGF-backed firms do not perform better than non-GGF VC-backed firms in terms of innovation expenditure and productivity. Furthermore, to shed light on whether the stage and amount GGFs invest in a firm matters in spurring firms' innovation activities, we repeat the regressions analogous to those in Table 12 where the main variable of interest is the *GGF_first* and *GGF_high* dummy. Table 13 Panel B reports the results. It shows that the coefficients of *GGF_first* and *GGF_high* dummy are all statistically insignificant regardless of which innovation activity measure is assessed, suggesting no variation in the effect of GGFs on firms' innovation expenditure or on innovation productivity across the stages and amounts GGFs invest in firms. Overall, this section documents no evidence implying that GGFs differ from other VC funds in promoting investee firms' innovation activities, regardless of the subsamples of GGF-backed firms assessed.

*****Insert Table 13 here*****

8. Conclusion

Using a sample of 2,855 firms that went public during the period between 2010 and 2021 in China, we examine the role of GGFs in the value creation for investee firms in the financial market and the product market. We show that GGF-backed firms enjoyed higher market valuation and lower IPO underpricing in comparison with non-VC-backed firms and even non-GGF VC-backed firms, indicating that GGFs could create values in the financial market both for the issuing firms through minimizing the 'money left on the table' and for stock investors through driving up the closing price on the first day of trading.

However, further evidence shows that the stock outperformance of GGF-backed firms at IPOs (i.e. manifested by their higher initial returns) did not persist, disappearing within two years after the IPOs. Consistently, evidence is also found that GGF-backed firms underperformed the non-GGF VC-backed firms and even the non-VC-backed firms in the product market post IPOs regardless of variables used to measure firm operating performance.

Moreover, we find no evidence that GGFs differ from other VCs in their effects on promoting investee firms' innovation expenditures or innovation productivity.

The results indicate that the higher initial returns of GGF-backed firms are more likely to arise from temporary market overvaluation rather than from the real benefits GGFs bring to their investee firms to improve their long-term performance. The evidence provides support to our conjecture that the average investor in China's stock market puts more weight on the benefits of political resources than on the concerns of capital misallocation and management inefficiency associated with GGF sponsoring when evaluating new shares.

The divergence of GGFs' value effects observed in the financial and the product market reveals the complexity of evaluating the value creation role of GGFs in Chinese market – while GGFs do not create real values in the product market for investee firms, they can spur investors' optimistic sentiment towards the firms which could create values (at least in the short term) in the financial market. The evidence also casts doubt on the ability of the investors at large to evaluate new shares, and highlights the crucial role of sentiment-driven trades in affecting stock valuation in the Chinese market. Following this line of thought, it justifies the efforts the Chinese regulators have made to promote the growth of institutional investors to stabilize the market such as the qualified foreign institutional investors (QFII) regime, and various policy initiatives to stimulate the growth the securities investment fund industry, etc.

In addition, given the increasingly important role of GGFs in China's VC market, the paper calls for attention from regulators to taking actions to address the problems of management inefficiency and possible capital misallocation associated with GGFs in order to bring real benefits to stimulate the long-term growth and innovation activities of their investee firms.

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Figure 1. Average abnormal monthly return post IPO for GGF-backed ('GGF'), non-GGF VC-backed ('NGGF') and non-VC-backed firms ('Non-VC').

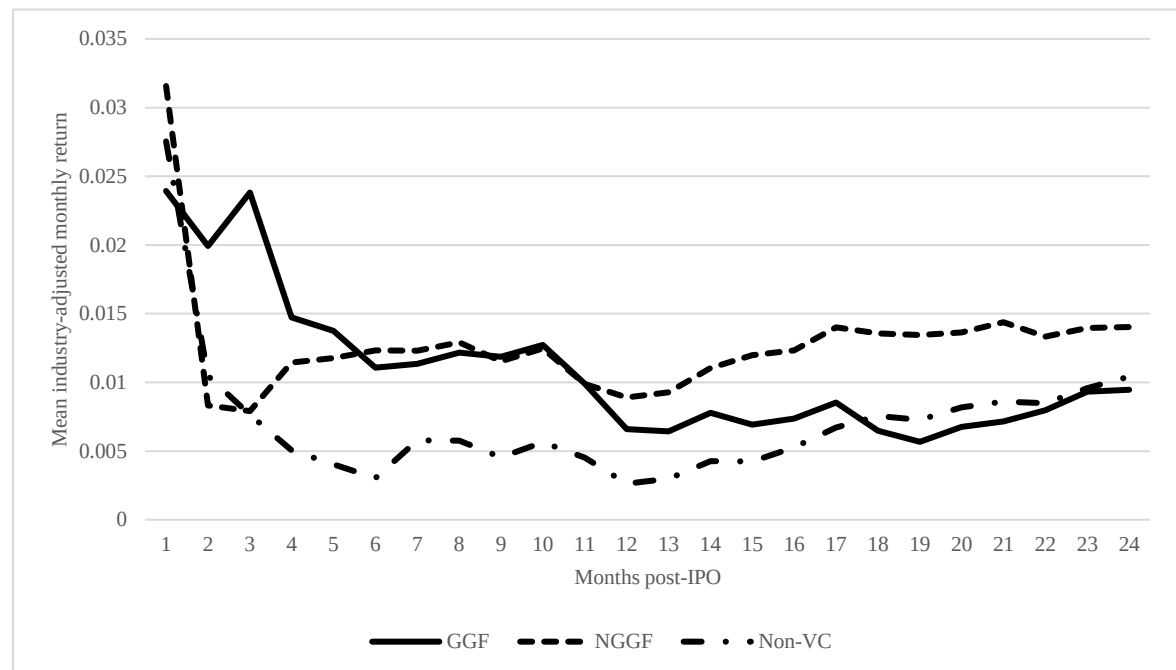


Table 1. Descriptive statistics of investments made by venture capital companies (left panel) and by GGF-backed venture capital companies (right panel)

	All VC investments					GGF-backed VC investments				
	# of inv	CNY of inv (bil CNY)	# of VC companies	#of investee firms	% of investee firm_IPO	# of inv	CNY of inv (bil CNY)	# of VC companies	#of investee firms	% of investee firm_IPO
Panel A. By year										
2010	141	1.52	96	74	0.622	8	0.05	7	8	0.375
2011	240	2.92	121	115	0.600	15	0.21	12	13	0.615
2012	234	3.01	124	119	0.597	24	0.25	16	20	0.700
2013	227	5.93	131	138	0.457	32	0.25	18	27	0.556
2014	417	6.24	188	259	0.212	49	0.45	31	47	0.298
2015	695	11.22	304	429	0.177	77	0.86	43	70	0.200
2016	952	12.50	449	610	0.180	118	1.03	66	104	0.231
2017	1,399	30.56	561	831	0.211	195	3.43	81	176	0.267
2018	2,242	46.55	805	1,352	0.137	317	6.54	123	279	0.222
2019	4,457	103.98	893	2,581	0.071	531	10.40	171	454	0.161
2020	4,970	115.39	949	2,452	0.039	848	12.66	200	650	0.077
2021	7,165	116.64	1,044	3,412	0.003	1,371	12.55	228	1,037	0.002
Total	23,139	456.44	2,704	9,059	0.069	3,585	48.69	376	2,484	0.100
Panel B. By industry										
IT	5,593	86.83	1,132	2,318	0.028	776	8.82	186	550	0.045
biotechnology and health	5,558	119.86	864	1,657	0.064	1,020	12.55	178	647	0.080
Semiconductor and electronic equipment	3,252	57.31	701	1,030	0.127	705	12.54	164	443	0.140
Internet	2,410	61.99	687	1,003	0.005	213	2.40	87	171	0.012
Machinery Manufacturing	1,144	18.11	464	501	0.172	205	2.13	103	151	0.166
Clean technology	634	11.47	291	253	0.198	112	1.59	69	84	0.310
Chemical materials	608	10.12	280	267	0.221	119	1.35	67	87	0.253
Media and Entertainment	577	5.04	214	410	0.015	51	0.71	26	40	0.075
Financial service	542	12.35	270	215	0.009	48	0.64	38	37	0.000
Retailing	530	12.13	200	273	0.029	63	0.79	39	51	0.078
Other	2,291	61.22	1,497	1,132	0.092	273	5.19	249	223	0.121

Table 2. Number of IPO firms in the sample by year.

Year	# of IPO firms	# of VC-backed IPO firms	# of GGF-backed IPO firms
2010	304	0	0
2011	277	1	0
2012	153	1	0
2013	0	0	0
2014	122	0	0
2015	216	2	0
2016	217	7	1
2017	431	17	3
2018	96	3	0
2019	187	43	11
2020	384	177	62
2021	468	208	96
Total	2,855	459	173

Table 3. Descriptive statistics for IPO firms.

	Mean	Median	Std Dev	p10	p90	Mean	Median	Std Dev	p10	p90	T-tests on the difference in means	
Panel A. VC-backed IPO firms versus non-VC-backed IPO firms	VC					Non-VC					VC-NVC	p-value
Return _{int}	1.461	0.761	1.754	0.431	3.331	0.598	0.440	0.842	0.081	1.092	0.863***	0.000
Underprc	0.628	0.622	1.061	-0.501	1.903	1.364	1.104	1.374	-0.014	3.011	-0.736***	0.000
Overval	0.833	0.552	2.044	-1.097	3.147	-0.766	-0.653	1.637	-2.544	0.818	1.598***	0.000
Age (yrs)	15.208	14.844	5.402	8.816	22.049	13.596	13.038	5.966	6.723	21.214	1.612***	0.000
Asset _{preIPO} (mil CNY)	209.491	168.305	230.088	85.769	333.600	248.141	125.180	1353.651	59.904	316.000	-38.65	0.542
EPS _{preIPO}	1.039	0.835	1.368	0.240	1.938	0.902	0.770	0.713	0.370	1.570	0.137***	0.002
SOE	0.035	0.000	0.184	0.000	0.000	0.091	0.000	0.288	0.000	0.000	-0.056***	0.000
Daygap	11.910	12.000	2.749	9.000	15.000	11.573	11.000	8.255	8.000	14.000	0.337	0.387
Value _{udw} (mil CNY)	72,810	33,137	96,930	1,841	192,398	55,892	29,130	75,398	2,036	139,304	16,918***	0.000
IPOhot	0.782	1.000	0.414	0.000	1.000	0.389	0.000	0.488	0.000	1.000	0.393***	0.000
No. of Firms	459					2,396						
Panel B. GGF-backed IPO firms versus non-GGF VC-backed IPO firms	GGF					Non-GGF					GGF-NGGF	p-value
Return _{int}	1.743	1.141	1.820	0.375	3.866	1.291	0.544	1.694	0.440	2.846	0.452***	0.007
Underprc	0.436	0.467	1.052	-0.636	1.559	0.745	0.709	1.051	-0.334	2.136	-0.309***	0.002
Overval	1.308	0.855	2.103	-0.698	3.929	0.547	0.363	1.957	-1.310	2.668	0.761***	0.000
Age (yrs)	15.116	14.878	5.224	8.816	21.041	15.263	14.814	5.514	8.847	22.704	-0.147	0.778
Asset _{preIPO} (mil)	198.351	176.677	163.799	86.510	311.591	216.190	159.483	262.054	84.000	349.580	-17.839	0.422
EPS _{preIPO}	0.965	0.840	0.982	0.210	1.900	1.083	0.825	1.555	0.250	2.030	-0.117	0.374
SOE	0.017	0.000	0.131	0.000	0.000	0.045	0.000	0.209	0.000	0.000	-0.028	0.114
Daygap	11.779	11.500	2.731	9.000	15.000	11.990	12.000	2.761	9.000	14.000	-0.21	0.428
Value _{udw} (mil)	80,336	34,109	104,268	2,946	259,779	68,284	31,783	92,129	1,621	167,373	12,052.15	0.198
IPOhot	0.831	1.000	0.375	0.000	1.000	0.752	1.000	0.433	0.000	1.000	0.080**	0.046
Age _{vc} (yrs)	9.034	8.275	4.981	3.614	15.684	8.163	7.066	5.771	1.988	16.693	0.871	0.116
#Fund _{vc}	49.718	33.179	44.768	7.200	115.250	38.355	26.000	39.416	4.000	99.500	11.363***	0.008
No. of Firms	173					286						

Notes: Return_{int} is IPO initial return, calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. Agevc is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fundvc is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 4. Effect of VC sponsoring on IPO initial return, underpricing and overvaluation.

	Panel A. All firms			Panel B. GGF-backed and non-VC-back firms			Panel C. Non-GGF VC-back and non-VC-back firms		
	Return _{int}	Underprc	Overval	Return _{int}	Underprc	Overval	Return _{int}	Underprc	Overval
VC	0.169* (0.094)	-0.257*** (0.000)	0.440*** (0.000)	0.368*** (0.008)	-0.418*** (0.000)	0.804*** (0.000)	0.063 (0.453)	-0.190** (0.013)	0.268** (0.018)
Age	0.017 (0.571)	0.100 (0.135)	-0.100 (0.252)	0.015 (0.689)	0.060 (0.273)	-0.055 (0.488)	0.024 (0.334)	0.086 (0.182)	-0.071 (0.378)
Asset _{preIPO}	-0.114*** (0.000)	0.161*** (0.000)	-0.274*** (0.000)	-0.108*** (0.000)	0.146*** (0.000)	-0.252*** (0.000)	-0.106*** (0.000)	0.164*** (0.000)	-0.273*** (0.000)
EPS _{preIPO}	-0.298*** (0.001)	0.210*** (0.000)	-0.519*** (0.000)	-0.267*** (0.004)	0.222*** (0.000)	-0.495*** (0.000)	-0.254*** (0.001)	0.217*** (0.000)	-0.478*** (0.000)
SOE	0.077 (0.130)	0.114 (0.132)	-0.035 (0.695)	0.114*** (0.007)	0.125* (0.076)	-0.004 (0.963)	0.072 (0.105)	0.112 (0.114)	-0.040 (0.613)
Daygap	0.002*** (0.003)	0.002* (0.061)	-0.000 (0.844)	0.002*** (0.010)	0.002 (0.121)	0.000 (0.948)	0.003*** (0.001)	0.002** (0.044)	0.001 (0.755)
ln(Value _{udw})	0.007*** (0.004)	-0.010* (0.057)	0.018*** (0.001)	0.006* (0.087)	-0.006 (0.331)	0.013** (0.034)	0.004* (0.098)	-0.009 (0.102)	0.013** (0.021)
IPOhot	0.235*** (0.002)	-0.104 (0.333)	0.331*** (0.000)	0.207*** (0.005)	-0.114 (0.304)	0.311*** (0.000)	0.237*** (0.001)	-0.133 (0.210)	0.366*** (0.000)
Constant	1.529*** (0.000)	-1.002** (0.048)	2.547*** (0.000)	1.555*** (0.000)	-0.787 (0.166)	2.343*** (0.001)	1.477*** (0.000)	-0.984* (0.059)	2.501*** (0.000)
P-value for SUE for the VC coefficient between the two subsamples							0.019		
	Return _{int}	Underprc	Overval				0.007		
							0.000		
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,855	2,849	2,849	2,569	2,563	2,563	2,682	2,677	2,677
Adj R ²	0.307	0.595	0.568	0.319	0.609	0.580	0.291	0.604	0.571

Notes: Panel A reports the results for all the sample IPO firms. Panel B reports the results for GGF-backed firms and non-VC-backed firms. Panel C reports the results for non-GGF VC-backed firms and non-VC-backed firms. Return_{int} is IPO initial return, calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. VC is a dummy indicating that a firm received VC investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 5. Effect of VC sponsoring on IPO initial return, underpricing and overvaluation, after EBM based on all independent variables except for VC dummy.

	Panel A. All firms			Panel B. GGF-backed and non-VC-back firms			Panel C. Non-GGF VC-back and non-VC-back firms		
	Return _{int}	Underprc	Overval	Return _{int}	Underprc	Overval	Return _{int}	Underprc	Overval
VC	0.194*	-0.252***	0.460***	0.378***	-0.345***	0.739***	0.089	-0.211***	0.313***
	(0.089)	(0.000)	(0.000)	(0.009)	(0.000)	(0.000)	(0.341)	(0.003)	(0.003)
Age	0.021	0.389***	-0.436***	0.017	0.367***	-0.411**	0.035	0.392***	-0.402***
	(0.808)	(0.000)	(0.000)	(0.926)	(0.000)	(0.031)	(0.450)	(0.001)	(0.002)
Asset _{preIPO}	-0.296***	0.133***	-0.372***	-0.364***	0.084*	-0.367***	-0.278***	0.173***	-0.403***
	(0.000)	(0.003)	(0.000)	(0.000)	(0.087)	(0.000)	(0.000)	(0.003)	(0.000)
EPS _{preIPO}	-0.459***	0.193***	-0.675***	-0.454***	0.228***	-0.695***	-0.408***	0.200***	-0.628***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
SOE	0.099	0.058	0.057	0.431**	0.065	0.396	0.009	0.073	-0.088
	(0.684)	(0.779)	(0.806)	(0.036)	(0.651)	(0.140)	(0.965)	(0.664)	(0.575)
Daygap	-0.007	-0.004	-0.003	-0.017	-0.017*	0.001	0.009	-0.002	0.012
	(0.709)	(0.544)	(0.898)	(0.585)	(0.098)	(0.970)	(0.605)	(0.739)	(0.527)
ln(Value _{udw})	0.017***	-0.014*	0.032***	0.021**	-0.003	0.024**	0.009	-0.011	0.022*
	(0.000)	(0.091)	(0.002)	(0.014)	(0.798)	(0.044)	(0.153)	(0.144)	(0.079)
IPOhot	0.376**	0.081	0.252**	0.276*	0.115	0.095	0.392***	0.024	0.328***
	(0.027)	(0.560)	(0.045)	(0.089)	(0.416)	(0.578)	(0.003)	(0.879)	(0.001)
Constant	5.030***	-0.244	4.957***	6.995***	1.614***	4.788***	4.607***	-0.680*	4.996***
	(0.000)	(0.503)	(0.000)	(0.000)	(0.007)	(0.000)	(0.000)	(0.071)	(0.000)
P-value for SUE for the VC coefficient between the two subsamples							0.025		
	Return _{int}						0.081		
	Underprc						0.003		
	Overval								
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ind dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,855	2,849	2,849	2,569	2,563	2,563	2,682	2,677	2,677
Adj R ²	0.186	0.414	0.360	0.200	0.438	0.365	0.171	0.434	0.358

Notes: Panel A reports the results for all the sample IPO firms. Panel B reports the results for GGF-backed firms and non-VC-backed firms. Panel C reports the results for non-GGF VC-backed firms and non-VC-backed firms. Return_{int} is IPO initial return, calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. VC is a dummy indicating that a firm received VC investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 6. Heckman two-stage estimation.

	Panel A. All firms				Panel B. GGF-backed and non-VC-back firms				Panel C. Non-GGF VC-back and non-VC-back firms			
	First-step Prob(VC)	Second-step results			First-step Prob(VC)	Second-step results			First-step Prob(VC)	Second-step results		
		Return _{int}	Underprc	Overval		Return _{int}	Underprc	Overval		Return _{int}	Underprc	Overval
VC		0.170* (0.081)	-0.264*** (0.000)	0.449*** (0.000)		0.355*** (0.010)	-0.404*** (0.000)	0.780*** (0.000)		0.065 (0.419)	-0.196** (0.013)	0.276** (0.014)
Ln(#firm_prov)	0.038** (0.017)				0.072*** (0.002)				0.031 (0.114)			
Age	0.351** (0.019)	-0.083 (0.316)	0.145 (0.217)	-0.257 (0.107)	0.240 (0.116)	-0.028 (0.632)	0.111 (0.122)	-0.149 (0.227)	0.355*** (0.010)	-0.012 (0.862)	0.162 (0.280)	-0.201 (0.242)
Asset _{preIPO}	0.296*** (0.000)	-0.197** (0.028)	0.221** (0.021)	-0.433*** (0.000)	0.280*** (0.000)	-0.182** (0.014)	0.207*** (0.001)	-0.394*** (0.000)	0.256*** (0.000)	-0.133* (0.060)	0.243** (0.021)	-0.398*** (0.002)
EPS _{preIPO}	0.071 (0.146)	-0.318*** (0.002)	0.210*** (0.000)	-0.538*** (0.000)	0.008 (0.872)	-0.268*** (0.009)	0.205*** (0.000)	-0.476*** (0.000)	0.093* (0.077)	-0.261*** (0.002)	0.226*** (0.000)	-0.496*** (0.000)
SOE	-0.707*** (0.000)	0.002*** (0.005)	0.002* (0.084)	-0.000 (0.887)	-0.952*** (0.000)	0.002** (0.026)	0.002*** (0.004)	-0.001 (0.672)	-0.517*** (0.005)	0.003*** (0.002)	0.002* (0.066)	0.001 (0.723)
Daygap		0.008*** (0.009)	-0.012** (0.039)	0.021*** (0.002)		0.008 (0.101)	-0.009 (0.203)	0.017** (0.027)		0.005* (0.089)	-0.011* (0.062)	0.016** (0.018)
Ln(Value _{udw})		0.276 (0.110)	0.002 (0.992)	0.313 (0.206)		0.349* (0.066)	0.003 (0.985)	0.367 (0.107)		0.120 (0.315)	-0.022 (0.914)	0.177 (0.417)
IPOhot		0.232*** (0.002)	-0.114 (0.272)	0.341*** (0.000)		0.204*** (0.008)	-0.142 (0.207)	0.339*** (0.000)		0.234*** (0.001)	-0.145 (0.157)	0.379*** (0.000)
IMR		-0.339 (0.219)	0.167 (0.684)	-0.560 (0.167)		-0.256 (0.172)	0.129 (0.555)	-0.394 (0.120)		-0.111 (0.631)	0.268 (0.602)	-0.449 (0.377)
Constant	-5.194*** (0.000)	3.164** (0.037)	-1.950 (0.299)	5.410** (0.015)	-4.909*** (0.000)	2.674** (0.021)	-2.276** (0.035)	5.032*** (0.004)	-4.811*** (0.000)	2.015 (0.101)	-2.371 (0.292)	4.776* (0.055)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies		Yes	Yes	Yes		Yes	Yes	Yes		Yes	Yes	Yes
Ind& Year-clst SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,827	2,827	2,822	2,822	2,395	2,395	2,390	2,390	2,653	2,653	2,649	2,649
Pseudo R ²	0.070				0.076				0.057			
Adj R ²		0.308	0.595	0.570		0.323	0.609	0.585		0.291	0.602	0.570

Notes: Panel A reports the results for all the sample IPO firms. Panel B reports the results based on GGF-backed firms and non-VC-backed firms. Panel C reports the results based on non-GGF VC-backed firms and non-VC-backed firms. Return_{int} is IPO initial return, calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. #firm_prov is number of firms invested by VCs in the province a given IPO firm is registered. VC is a dummy indicating that a firm received VC investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 7. Effect of GGF sponsoring on IPO initial return, underpricing and overvaluation based on the sample of VC-backed IPO firms, after EBM based on all independent variables except for GGF dummy.

	Panel A.			Panel B.		
	Return _{int}	Underprc	Overval	Return _{int}	Underprc	Overval
GGF	0.161* (0.076)	-0.149 (0.149)	0.299*** (0.004)	0.158* (0.051)	-0.106 (0.297)	0.243** (0.037)
Age	0.067 (0.487)	0.379*** (0.000)	-0.502*** (0.006)	-0.009 (0.950)	0.455*** (0.000)	-0.703** (0.010)
Asset _{preIPO}	-0.220*** (0.000)	0.118* (0.090)	-0.294** (0.014)	-0.184** (0.012)	0.140 (0.123)	-0.272* (0.087)
EPS _{preIPO}	-0.555*** (0.000)	0.089** (0.026)	-0.693*** (0.000)	-0.566*** (0.000)	0.078 (0.224)	-0.705*** (0.000)
Daygap	-0.029* (0.077)	0.001 (0.967)	-0.031 (0.248)	-0.035* (0.071)	0.003 (0.818)	-0.040 (0.214)
ln(Value _{udw})	0.024** (0.015)	-0.025*** (0.004)	0.055*** (0.000)	0.023* (0.068)	-0.023** (0.012)	0.053*** (0.008)
SOE	0.445 (0.443)	0.051 (0.886)	0.562 (0.382)	0.016 (0.979)	-0.034 (0.882)	0.143 (0.824)
Age _{vc}				0.007 (0.672)	-0.019* (0.056)	0.025 (0.227)
#Fund _{vc}				0.000 (0.939)	-0.000 (0.577)	0.000 (0.944)
IPOhot	0.407 (0.241)	0.179 (0.254)	0.249 (0.348)	0.445 (0.268)	0.136 (0.370)	0.336 (0.240)
Constant	2.516*** (0.000)	-1.603** (0.033)	4.325*** (0.000)	2.941*** (0.001)	1.742** (0.034)	1.658 (0.262)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	459	458	458	382	381	381
Adj R ²	0.218	0.288	0.305	0.263	0.311	0.343

Notes: Return_{int} is IPO initial return, calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. GGF is a dummy indicating that a firm received GGF investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. Age_{vc} is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fund_{vc} is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 8. Heterogeneity test by the stage GGFs make investments- Effect of GGF sponsoring on IPO initial return, underpricing and overvaluation, after EBM based on all independent variables except for GGF dummy.

	Panel A. GGF-first			Panel B. GGF-late		
	Return _{int}	Underprc	Overval	Return _{int}	Underprc	Overval
GGF	-0.026 (0.804)	-0.114 (0.102)	0.069 (0.520)	0.362*** (0.000)	-0.099 (0.552)	0.447* (0.070)
Age	0.126 (0.658)	0.219*** (0.005)	-0.141 (0.643)	-0.040 (0.829)	0.440** (0.023)	-0.734** (0.029)
Asset _{preIPO}	-0.077 (0.585)	0.066 (0.365)	-0.107 (0.575)	-0.432*** (0.005)	0.101 (0.148)	-0.471** (0.023)
EPS _{preIPO}	-0.560*** (0.000)	0.036 (0.352)	-0.623*** (0.000)	-0.363*** (0.002)	0.242*** (0.009)	-0.763*** (0.000)
Daygap	-0.013 (0.429)	0.014 (0.142)	-0.032 (0.139)	-0.047 (0.136)	-0.010 (0.431)	-0.040 (0.152)
ln(Value _{udw})	-0.010 (0.540)	-0.018* (0.063)	0.009 (0.315)	0.072** (0.037)	0.005 (0.699)	0.067* (0.058)
SOE	-0.061 (0.753)	-0.711* (0.062)	0.938* (0.092)	0.497 (0.509)	0.214 (0.591)	0.310 (0.723)
Age _{vc}	0.044*** (0.002)	-0.019 (0.117)	0.061*** (0.005)	0.009 (0.748)	-0.038*** (0.000)	0.046* (0.052)
#Fund _{vc}	-0.002*** (0.000)	0.001 (0.384)	-0.004*** (0.010)	0.000 (0.980)	-0.001 (0.442)	0.001 (0.892)
IPOhot	0.458 (0.152)	-0.156 (0.237)	0.633** (0.011)	0.366 (0.284)	0.111 (0.453)	0.301 (0.247)
Constant	1.251 (0.277)	3.279*** (0.001)	-2.089 (0.225)	5.674*** (0.000)	-1.319* (0.081)	7.669*** (0.000)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	281	281	281	320	319	319
Adj R ²	0.298	0.456	0.432	0.340	0.305	0.398

Notes: Panel A reports the results based on the firms that received GGF investments as their first VC investments and non-GGF VC-backed firms. Panel B reports the results based on the firms that received GGF investments at later stage and non-GGF VC-backed firms. Return_{int} is IPO initial return, calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. GGF is a dummy indicating that a firm received GGF investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. Age_{vc} is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fund_{vc} is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 9. Heterogeneity test by the amount of GGF investments- Effect of GGF sponsoring on IPO initial return, underpricing and overvaluation, after EBM based on all independent variables except for GGF dummy.

	Panel A. GGF-high			Panel B. GGF-low		
	Return _{int}	Underprc	Overval	Return _{int}	Underprc	Overval
GGF	-0.246*** (0.007)	-0.056 (0.489)	-0.108 (0.326)	0.387*** (0.005)	-0.041 (0.794)	0.364*** (0.009)
Age	-0.185 (0.275)	0.415** (0.020)	-0.838*** (0.001)	0.053 (0.676)	0.421*** (0.002)	-0.612** (0.026)
Asset _{preIPO}	-0.006 (0.946)	0.190 (0.156)	-0.038 (0.833)	-0.230*** (0.005)	0.184* (0.055)	-0.397** (0.014)
EPS _{preIPO}	-0.460*** (0.000)	0.025 (0.782)	-0.661*** (0.000)	-0.375*** (0.000)	0.167*** (0.001)	-0.658*** (0.000)
Daygap	-0.020 (0.178)	0.019 (0.419)	-0.032 (0.287)	-0.043* (0.054)	0.013 (0.449)	-0.061* (0.078)
ln(Value _{udw})	0.035** (0.031)	-0.028** (0.031)	0.077*** (0.001)	-0.000 (0.995)	-0.023 (0.228)	0.024 (0.255)
SOE	-0.275 (0.618)	-0.532* (0.076)	0.191 (0.819)	0.076 (0.933)	0.229* (0.079)	-0.137 (0.879)
Agevc	-0.001 (0.964)	-0.012 (0.297)	0.014 (0.547)	0.012 (0.330)	-0.027** (0.032)	0.036 (0.142)
#Fundvc	-0.002 (0.556)	0.001 (0.462)	-0.004* (0.081)	0.001 (0.700)	-0.000 (0.767)	0.002 (0.713)
IPOhot	0.774*** (0.009)	-0.120 (0.654)	0.942** (0.024)	0.443 (0.332)	0.293 (0.136)	0.238 (0.388)
Constant	0.743 (0.542)	-0.500 (0.755)	0.920 (0.628)	2.280*** (0.000)	0.898 (0.342)	2.490 (0.149)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Ind dummies	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	267	266	266	331	331	331
Adj R ²	0.295	0.279	0.364	0.198	0.350	0.347

Notes: Panel A reports the results based on the firms for which the GGF investments account for more than 50% of the total VC investments a firm received prior to IPOs and non-GGF VC-backed firms. Panel B reports the results based on the firms for which the GGF investments account for less than or equal to 50% of the total VC investments a firm received prior to IPOs and non-GGF VC-backed firms. Return_{int} is IPO initial return, calculated as the ratio of the difference between the first-day closing price and the IPO offer price to the offer price. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. GGF is a dummy indicating that a firm received GGF investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. Agevc is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fundvc is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 10. Effect of VC sponsoring on firms' stock performance within two years post IPO

	Panel A. After EBM		Panel B. GGF heterogeneity	
	(i)	(ii)	(i)	(ii)
VC	0.002 (0.569)	0.005** (0.037)		
GGF_first			-0.019** (0.026)	
GGF_high				-0.002 (0.282)
Age	0.008*** (0.000)	0.001 (0.850)	0.015*** (0.000)	0.011** (0.025)
Asset _{preIPO}	0.001 (0.464)	0.000 (0.971)	0.002 (0.476)	0.005*** (0.007)
EPS _{preIPO}	0.006*** (0.000)	0.003*** (0.006)	0.006*** (0.000)	0.006*** (0.001)
SOE	-0.002 (0.722)	0.013 (0.290)	-0.018*** (0.000)	-0.008*** (0.000)
Age _{vc}			-0.002*** (0.000)	-0.002*** (0.000)
#Fund _{vc}			0.000 (0.490)	0.000 (0.295)
Constant	-0.069*** (0.000)	-0.019 (0.611)	0.107** (0.028)	0.073*** (0.001)
Year dummies	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes
Observations	2,213	2,310	73	71
Adj R ²	0.202	0.048	0.396	0.341

Notes: Panel A specification (i) reports the results based on GGF-backed firms and non-VC-backed firms after EBM on all the independent variables except for VC dummy, specification (ii) reports the results based on non-GGF VC-backed firms and non-VC-backed firms after EBM on all the independent variables except for VC dummy. Panel B reports the results based on GGF-backed firms. The dependent variable is a firm's average abnormal return (relative to the median of the industry) over the 24 months post IPOs. VC is a dummy indicating that a firm received VC investments prior to IPOs. GGF_first is a dummy indicating that a GGF-backed firm received the GGF investment as its first VC investment. GGF_high is a dummy indicating that GGF investments account for more than 50% of the total VC investments a firm received prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Age_{vc} is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fund_{vc} is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 11. Effect of VC sponsoring on firms' operating performance within two years post IPO

	Panel A. GGF versus non-VC-back			Panel B. Non-GGF versus non-VC-back			Panel C. GGF versus non-GGF VC-back		
	ROA	ROE	EBIT/asset	ROA	ROE	EBIT/asset	ROA	ROE	EBIT/asset
VC	-0.006*	-0.016***	-0.009*	0.002	0.000	0.003			
	(0.088)	(0.004)	(0.087)	(0.572)	(0.969)	(0.541)			
GGF							-0.018**	-0.027***	-0.024**
							(0.018)	(0.009)	(0.019)
Age	0.022***	0.032***	0.027***	0.008*	0.013*	0.009*	0.026***	0.040***	0.032***
	(0.000)	(0.000)	(0.000)	(0.055)	(0.089)	(0.073)	(0.000)	(0.000)	(0.000)
Asset _{preIPO}	-0.007***	-0.018***	-0.008***	-0.007***	-0.011***	-0.007***	-0.021***	-0.023***	-0.021***
	(0.002)	(0.002)	(0.000)	(0.001)	(0.000)	(0.002)	(0.000)	(0.000)	(0.000)
EPS _{preIPO}	0.029***	0.028***	0.031***	0.017***	0.015***	0.018***	0.024***	0.026***	0.025***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
SOE	0.001	0.007	-0.000	0.006**	0.014**	0.007**	-0.015***	-0.022***	-0.020***
	(0.696)	(0.312)	(0.897)	(0.013)	(0.015)	(0.033)	(0.000)	(0.000)	(0.000)
Age _{vc}							-0.001**	-0.001***	-0.001**
							(0.027)	(0.001)	(0.042)
#Fund _{vc}							0.000	0.000	0.000
							(0.907)	(0.188)	(0.637)
Constant	0.064*	0.169***	0.051	0.051*	0.088***	0.045	0.228***	0.267***	0.234***
	(0.081)	(0.000)	(0.101)	(0.082)	(0.008)	(0.142)	(0.000)	(0.000)	(0.000)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,213	2,212	2,211	2,310	2,309	2,308	208	208	208
Adj R ²	0.297	0.259	0.298	0.091	0.056	0.085	0.266	0.265	0.261

Notes: Panel A reports the results based on GGF-backed firms and non-VC-backed firms. Panel B reports the results based on non-GGF VC-backed firms and non-VC-backed firms. Panel C reports the results based on VC-backed firms. The column name indicates the dependent variable in each specification: ROA is the average annual return on assets over two years post IPOs; ROE is the average annual return on equity over two years post IPOs; EBIT/asset is the average annual ratio of EBIT to total assets over two years post IPOs. VC is a dummy indicating that a firm received VC investments prior to IPOs. GGF is a dummy indicating that a firm received GGF investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Age_{vc} is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fund_{vc} is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively

Table 12. Heterogeneity test by the stage GGFs make investments (Panel A) and by the amount of GGF investments (Panel B) - Effect of GGF sponsoring on firms' operating performance within two years post IPO.

	Panel A			Panel B		
	ROA	ROE	EBIT/asset	ROA	ROE	EBIT/asset
GGF_first	-0.025* (0.064)	-0.042** (0.020)	-0.028* (0.059)			
GGF_high				-0.016* (0.062)	-0.015 (0.123)	-0.013 (0.155)
Age	0.032** (0.046)	0.040** (0.028)	0.038** (0.041)	0.033** (0.010)	0.034** (0.033)	0.035** (0.030)
Asset _{preIPO}	-0.040*** (0.001)	-0.047*** (0.010)	-0.041*** (0.001)	-0.038*** (0.001)	-0.042** (0.020)	-0.038*** (0.002)
EPS _{preIPO}	0.025*** (0.000)	0.029*** (0.000)	0.026*** (0.000)	0.022*** (0.000)	0.026*** (0.000)	0.024*** (0.000)
SOE	-0.050** (0.015)	-0.051* (0.081)	-0.056** (0.014)	-0.025** (0.018)	-0.020* (0.084)	-0.033*** (0.007)
Age _{vc}	-0.001 (0.258)	-0.001 (0.376)	-0.001 (0.308)	-0.001 (0.468)	-0.001 (0.514)	-0.001 (0.473)
#Fund _{vc}	-0.000 (0.427)	-0.000 (0.276)	-0.000 (0.407)	-0.000 (0.405)	-0.000 (0.549)	-0.000 (0.446)
Constant	0.380*** (0.000)	0.482*** (0.002)	0.387*** (0.000)	0.372*** (0.000)	0.444*** (0.004)	0.368*** (0.000)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	73	73	73	71	71	71
Adj R ²	0.587	0.570	0.592	0.551	0.499	0.550

Notes: The column name indicates the dependent variable in each specification: ROA is the average annual return on assets over two years post IPOs; ROE is the average annual return on equity over two years post IPOs; EBIT/asset is the average annual ratio of EBIT to total assets over two years post IPOs. GGF_first is a dummy indicating that a GGF-backed firm received the GGF investment as its first VC investment. GGF_high is a dummy indicating that GGF investments account for more than 50% of the total VC investments a firm received prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Age_{vc} is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fund_{vc} is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table 13. Effect of GGF sponsoring on firms' innovation activities within two years post IPO.

	Panel A. GGF-back versus non-GGF VC-back			Panel B. Heterogeneity in GGF-back group					
	(i) RD/revenue	(ii) ln(1+Aplpt)	(iii) ln(1+Obtpt)	(i) RD/revenue	(ii) ln(1+Aplpt)	(iii) ln(1+Obtpt)	(iv) RD/revenue	(v) ln(1+Aplpt)	(vi) ln(1+Obtpt)
GGF	2.885 (0.169)	0.302 (0.268)	0.077 (0.695)						
GGF_first				-0.389 (0.766)	0.523 (0.240)	0.123 (0.643)			
GGF_high							-0.252 (0.832)	0.130 (0.769)	0.449 (0.246)
Age	-3.532*** (0.001)	-1.142*** (0.003)	-0.617*** (0.000)	-4.672** (0.013)	-1.415*** (0.006)	-0.695* (0.100)	-4.157** (0.018)	-1.682*** (0.000)	-0.830 (0.115)
Asset _{preIPO}	-0.391 (0.756)	-0.044 (0.766)	0.076 (0.393)	1.491 (0.347)	0.280 (0.388)	0.120 (0.633)	1.536 (0.354)	0.207 (0.508)	0.140 (0.539)
EPS _{preIPO}	-1.856*** (0.005)	-0.283*** (0.001)	-0.333*** (0.000)	-1.846*** (0.008)	-0.403*** (0.005)	-0.462*** (0.000)	-1.944** (0.023)	-0.339*** (0.003)	-0.411*** (0.000)
SOE	-2.037* (0.060)	0.913 (0.247)	0.608 (0.275)	0.650 (0.643)	2.794*** (0.000)	2.247*** (0.000)	0.556 (0.658)	2.695*** (0.000)	2.371*** (0.000)
Age _{vc}	0.323** (0.019)	0.065*** (0.000)	0.046*** (0.000)	0.206 (0.141)	0.068*** (0.003)	0.023 (0.297)	0.206 (0.142)	0.067** (0.049)	0.015 (0.600)
#Fund _{vc}	-0.035*** (0.000)	-0.008*** (0.000)	-0.006*** (0.002)	-0.026*** (0.008)	-0.005* (0.059)	-0.003 (0.299)	-0.026** (0.026)	-0.006** (0.028)	-0.002 (0.341)
Constant	21.595 (0.186)	3.935* (0.069)	1.708 (0.126)	9.114 (0.552)	1.119 (0.787)	1.392 (0.564)	7.167 (0.623)	2.932 (0.369)	1.552 (0.471)
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	205	206	206	73	73	73	71	71	71
Adj R ²	0.284	0.153	0.124	0.093	0.114	0.078	0.081	0.119	0.066

Notes: Panel A reports the results based on VC-backed firms. Panel B reports the results based on GGF-backed firms. The column name indicates the dependent variable in each specification: RD/revenue is a firm's average yearly ratio of R&D expenditure to total revenue within two years post IPOs; Aplpt is the average number of patents applied for per year within two years post IPOs; Obtpt is the average number of patents obtained per year within two years post IPOs. GGF is a dummy indicating that a firm received GGF investments prior to IPOs. GGF_first is a dummy indicating that a GGF-backed firm received the GGF investment as its first VC investment. GGF_high is a dummy indicating that GGF investments account for more than 50% of the total VC investments a firm received prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Age_{vc} is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fund_{vc} is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively

Appendix

Table A.1. Effect of GGF sponsoring on the probability of firms going public.

	(i)	(ii)
GGF	0.039*** (0.000)	0.028*** (0.005)
Value _{inv}		0.000* (0.051)
#Fund _{inv}		0.005 (0.129)
Age _{vc}		-0.008*** (0.000)
#Fund _{vc}		0.001*** (0.001)
Constant	0.057*** (0.002)	0.096*** (0.006)
Province dummies	No	Yes
Industry clustered SE	Yes	Yes
Observations	9,041	4,546
Adj R ²	0.005	0.083

Notes: The dependent variable is a dummy indicating a firm went public as at the end of 2021. GGF is a dummy indicating that a firm received GGF investments as of 2021 (for the firms that did not go public) or prior to IPOs (for the firms that went public). Value_{inv} is the total CNY amount of VC investments a firm received prior to the IPO. #Fund_{inv} is the number of VC funds making investments prior to the IPOs. Age_{VC} is the average number of years a VC company had been in operation till the month an investment was made across all the investments a given firm received prior to the IPO. #Fund_{VC} is the average number of funds a VC company had managed till the month an investment was made across all the investments a given firm received prior to the IPO. Province dummies indicate the province a given firm is registered in. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table A.2. Analogous to Table 4. Effect of VC sponsoring on IPO initial return, underpricing and overvaluation, when a firm's intrinsic value is estimated using the mean P/E ratio of the same industry it comes from.

	Panel A. All firms		Panel B. GGF-backed and non-VC-back firms		Panel C. Non-GGF VC-back and non-VC-back firms	
	Underprc	Overval	Underprc	Overval	Underprc	Overval
VC	-0.497*** (0.000)	0.698*** (0.000)	-0.698*** (0.000)	1.096*** (0.000)	-0.420** (0.011)	0.525*** (0.004)
Age	0.161 (0.271)	-0.171 (0.314)	0.095 (0.453)	-0.097 (0.521)	0.134 (0.338)	-0.126 (0.427)
Asset _{preIPO}	0.454*** (0.000)	-0.566*** (0.000)	0.423*** (0.000)	-0.524*** (0.000)	0.465*** (0.000)	-0.573*** (0.000)
EPS _{preIPO}	0.463*** (0.000)	-0.791*** (0.000)	0.475*** (0.000)	-0.759*** (0.000)	0.483*** (0.000)	-0.760*** (0.000)
SOE	0.199 (0.255)	-0.126 (0.481)	0.209 (0.204)	-0.083 (0.641)	0.205 (0.208)	-0.138 (0.405)
Daygap	0.002 (0.386)	-0.000 (0.895)	0.001 (0.600)	0.000 (0.907)	0.002 (0.398)	0.001 (0.862)
ln(Value _{udw})	-0.032*** (0.007)	0.038*** (0.002)	-0.022 (0.119)	0.026* (0.063)	-0.031** (0.015)	0.033*** (0.008)
IPOhot	-0.243 (0.361)	0.468** (0.028)	-0.257 (0.356)	0.451* (0.050)	-0.297 (0.267)	0.529** (0.014)
Constant	-1.729 (0.108)	3.311*** (0.007)	-1.376 (0.250)	2.940** (0.022)	-1.699 (0.126)	3.246** (0.010)
P-value for SUE for the VC coefficient between the two subsamples					0.175 0.015	
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Ind dummies	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,849	2,849	2,849	2,849	2,677	2,677
Adj R ²	0.580	0.599	0.580	0.599	0.591	0.603

Notes: Panel A reports the results for all the sample IPO firms. Panel B reports the results for GGF-backed firms and non-VC-backed firms. Panel C reports the results for non-GGF VC-backed firms and non-VC-backed firms. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. VC is a dummy indicating that a firm received VC investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.

Table A.3. Analogous to Table 5. Effect of VC sponsoring on IPO initial return, underpricing and overvaluation after EB, when a firm's intrinsic value is estimated using the mean P/E ratio of the same industry it comes from.

	Panel A. All firms		Panel B. GGF-backed and non-VC-back firms		Panel C. Non-GGF VC-back and non-VC-back firms	
	Underprc	Overval	Underprc	Overval	Underprc	Overval
VC	-0.442*** (0.000)	0.707*** (0.000)	-0.587*** (0.000)	1.023*** (0.000)	-0.387** (0.021)	0.539*** (0.003)
Age	0.715*** (0.000)	-0.739*** (0.000)	0.758*** (0.000)	-0.774*** (0.003)	0.762*** (0.003)	-0.729*** (0.005)
Asset _{preIPO}	0.310*** (0.000)	-0.522*** (0.000)	0.240*** (0.002)	-0.451*** (0.000)	0.370*** (0.000)	-0.575*** (0.000)
EPS _{preIPO}	0.451*** (0.000)	-1.000*** (0.000)	0.481*** (0.000)	-1.010*** (0.000)	0.506*** (0.000)	-0.992*** (0.000)
SOE	0.193 (0.676)	-0.111 (0.806)	0.015 (0.957)	0.430 (0.262)	0.274 (0.506)	-0.304 (0.423)
Daygap	-0.000 (0.964)	-0.001 (0.936)	-0.009 (0.331)	0.004 (0.742)	0.001 (0.855)	0.003 (0.750)
ln(Value _{udw})	-0.038 (0.149)	0.057** (0.036)	-0.011 (0.705)	0.023 (0.434)	-0.032 (0.126)	0.042** (0.018)
IPOhot	0.236 (0.551)	0.125 (0.653)	0.401 (0.292)	-0.161 (0.587)	0.164 (0.715)	0.235 (0.524)
Constant	-0.449 (0.617)	3.611*** (0.001)	1.486 (0.119)	2.730*** (0.007)	-1.127 (0.173)	4.117*** (0.001)
P-value for SUE for the VC coefficient between the two subsamples					0.293 0.028	
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
Ind dummies	Yes	Yes	Yes	Yes	Yes	Yes
Exg dummies	Yes	Yes	Yes	Yes	Yes	Yes
Year&Ind-clst SE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	2,849	2,849	2,563	2,563	2,677	2,677
Adj R ²	0.379	0.412	0.400	0.430	0.401	0.424

Notes: Panel A reports the results for all the sample IPO firms. Panel B reports the results for GGF-backed firms and non-VC-backed firms. Panel C reports the results for non-GGF VC-backed firms and non-VC-backed firms. Underprc is IPO underpricing, calculated as the difference between a firm's intrinsic value and offer price divided by the offer price. Overval is market overvaluation, calculated as the difference between the first-day closing price and the intrinsic value divided by the offer price. VC is a dummy indicating that a firm received VC investments prior to IPOs. Age is defined as the number of years an IPO firm has been in operation till the IPO date. Asset_{preIPO} and EPS_{preIPO} is the value of total assets and the value of the EPS disclosed in the prospectus prior to the IPO, respectively. SOE is a dummy indicating that an IPO firm is a state-owned firm. Daygap is defined as the number of days between the offering and the listing dates. Value_{udw} is the sum of the CNY amount of IPOs for which the lead underwriter had been the lead underwriter prior to the current IPO. IPOhot is a dummy indicating the 'hot' IPO period, i.e. when the median initial return across all the IPOs for a given month is larger than the median initial return across all the IPOs over the whole sample period of 2010-2021. P-values are reported in parentheses. ***, **, * are 1, 5 and 10 percent statistical significance respectively.